Final Year Project

emoBot

Mid - 1 Report

FYP Team

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Introduction:

This document is the Mid-1 report for the Final Year Project called emoBot. The document includes the SRS documentation for the project. Incorporating of all of the require diagrams and information. A little about the emoBot, we are developing an AI-driven Mental Health Support app. The app will use emotion recognition, with voice synthesis to deliver empathetic responses to the users in the form of either text or voice as the user intended.

Project Vision:

1. Project Problem:

Mental health issues are a critical and growing concern all around the globe and many mental health solution out their do not provide the responses that are empathetic toward the user and if they are they are text form. We are working to bridge this gap between user and text by introducing responses in the voice form using voice synthesis.

2. Objective:

- Develop an Ai chatbot that engages in the supportive and empathetic conversations.
- Integrating emotional detection and STT and TTS with Chatbot.
- Provide Content and CBT tools to the user.
- Provide crisis support if the need arrives by detection technology.

3. Scope:

4. Constraint:

These are the following constraint faced in the making of project:

- Limited access of the real-world mental health data.
- Time limitation for development within the academic year.
- Restriction on the use of external APIs.

5. Stakeholder:

- ### Stakeholder Summary:

Following are the stakeholders associated with the app.

End – Users: People who would interact with chatbot.

Admin: Person who will manage users and chats.

Expert: Person who will provide professional help in times of need.

Developer: Responsible for designing, developing, and maintaining the app.

- ### Key High Goals of Stakeholder:

  End – Users: People seeking Mental health support would intract with the application.

  Admin: Person who will manage users and chats to maintain the application.

  Expert: Person who will provide professional help in times of need.

  Developer: Responsible for designing, developing, and maintaining the app and for the completion of the Project.

Software Requirement Specification (SRS):

1. List of Features:

Our app is designed to offer empathetic conversations to users. It will use emotion recognition to generate responses that align with the user's emotional state. The key features include:

- Speech recognition and text-to-speech functionalities for seamless interactions.
- Mood tracking and journaling tools to help users monitor their emotional journey.
- CBT (Cognitive Behavioral Therapy) tools that provide users with exercises to manage mental health.
- Crisis detection capabilities to offer support when the user is in distress.
- A user management system that lets users create profiles and manage their experience.
- Personalized content and recommendations tailored to the user's emotional state.
- A feedback system where users can share their thoughts to help improve the app.

2. Functional Requirements:

These are the features our application will have to provide the users.

  - User Registration/Login: Users can create accounts, log in, and manage their profiles.
  - AI Chatbot Conversations: The chatbot engages users with empathetic responses after analyzing their emotions.
  - Emotion Recognition: The system will detect the user's emotions, whether through text or voice, and adjust its responses accordingly.

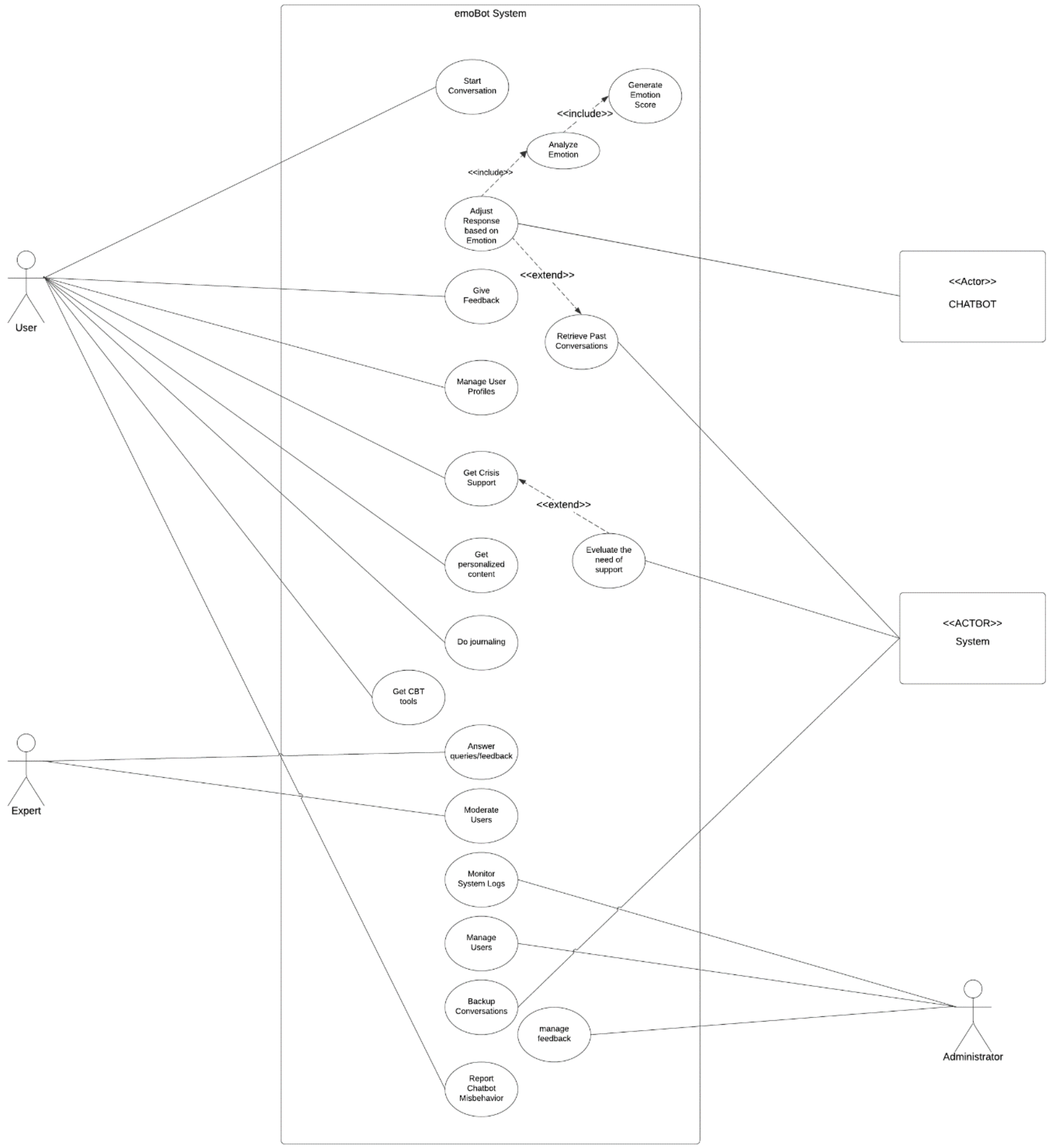
- Mood Tracking: Users will be able to log their moods over time, creating a personalized emotional history.
- CBT Tools: The app will provide therapeutic tools based on CBT to help users with mental health exercises.
- Crisis Support: The system will recognize signs of distress and offer appropriate support resources when necessary.
- Feedback Management: Users will be able to submit feedback that the admin team can review and address.
- Personalized Content: Content recommendations will be based on the user's behavior and emotional state for a more tailored experience.

3. Non-Functional Requirements:

These are the non-functional requirements of our application:

- **Data Privacy:** We'll ensure that all data is handled in line with privacy regulations like GDPR, prioritizing sensitive user data.
- **Response Time:** The chatbot will respond to user input within 1-2 seconds to ensure a smooth user experience.
- **Availability:** The system will be up and running 99.9% of the time, ensuring users can access support whenever they need it.
- **Accessibility:** The app will be accessible to users with disabilities, offering features like screen readers and voice commands.
- **Integration with External APIs:** We will integrate third-party APIs to support emotion recognition, speech-to-text, and text-to-speech functionalities, enhancing the app's capabilities.

Use Case Diagram:



High level Use Cases:

1. Start Conversation

Actor	User
Type	Primary
Description	The user initiates a conversation with the chatbot, which provides empathetic responses based on emotion detection and mood analysis.

2. Give Feedback

Actor	User
Type	Secondary
Description	The user provides feedback about their experience with the chatbot, which is stored for review by the admin and used to improve the system.

3. User Journal

Actor	User
Type	Primary
Description	User will be able to journal their thoughts about their daily lives.

4. Manage User Profiles

Actor	User
Type	Primary
Description	The user can update their profile information, including personal details, preferences, and mood tracking, to ensure personalized content and experiences.

5. Get Personalized Content

Actor	User
Type	Primary
Description	Based on the user's mood and behavior, the system provides personalized mental health content such as articles, exercises, or recommendations.

6. Access CBT Tools

Actor	User
Type	Primary

Description	The user can access various Cognitive Behavioral Therapy (CBT) tools designed to assist with anxiety, depression, and other mental health challenges.

7. Report Chatbot Misbehavior

Actor	User
Type	Secondary
Description	The user can report inappropriate or inaccurate responses from the chatbot, triggering a review process by the admin or expert.

8. Manage Feedback

Actor	Admin
Type	Primary
Description	The admin reviews and manages user feedback, making necessary changes to the chatbot or system based on the submitted feedback.

9. Manage Users

Actor	Admin
Type	Primary
Description	The admin manages user accounts, including creating, updating, suspending, or deleting users as necessary.

10. Moderate Users

Actor	Expert
Type	Secondary
Description	The expert moderates user interactions by reviewing flagged conversations and determining whether user behavior violates guidelines.

11. Answer Queries/Feedback

Actor	Expert
Type	Primary
Description	The expert provides direct responses to user queries or feedback that require human intervention.

12. Evaluate Need for Crisis Support

Actor	System
Type	Primary

Description	The system monitors conversations for signs of emotional distress and initiates crisis support protocols if a user is in danger.

13. Backup Conversations

Actor	System
Type	Secondary
Description	The system regularly backs up user conversations to prevent data loss and ensure recoverability.

14. Respond Based on Emotion

Actor	Chatbot
Type	Primary
Description	The chatbot analyzes the emotional state of the user, generates an emotion score, and provides empathetic responses based on the detected emotion.

Extended Use Cases:

1. Start Conversations

Use case Name	Start Conversations.	
Scope	User starts a conversation with the chatbot.	
Level	Primary	
Primary Actor	User	
Stakeholders and Interests	**User:** Needs empathetic responses and guidance. **Admin**: Ensures the conversation is safe and appropriate.	
Precondition	User is logged in.	
Postcondition	Ensures the conversation is safe and appropriate.	
	Actor Action	**System Requirements**

Main Success Scenario	1. User types or sends a voice message.	2. the chatbot responds appropriately based on emotion analysis.
Extensions	If the user expresses distress, the system evaluates the need for crisis support.	
Special Requirements	1. Emotion analysis model should be accurate and fast. 2. Latency of speech response should be fast and empathetic.	

2. Give Feedback

Usecase Name	Give Feedback	
Scope	User provides feedback on the chatbot or the app experience.	
Level	Primary	
Primary Actor	User	
Stakeholders and Interests	**User**: Wants to improve the chatbot experience or report an issue. **Admin**: Receives feedback to improve the system.	
Precondition	1. User is logged In. 2. User has interacted with the chatbot.	
Postcondition	Feedback is submitted and logged.	
Main Success Scenario	**Actor Action**	**System Requirements**
	1. User provides feedback through a feedback form.	2. Feedback is then stored in the database by the system.
Extensions	If feedback is negative, it triggers an alert for admin review.	
Special Requirements	Feedback form should be easy to access and use.	

3. Manage User Feedback

Usecase Name	Manage User Profiles
Scope	User manages their personal details, including updating information and setting preferences.
Level	Primary
Primary Actor	User

Stakeholders and Interests	**User**: Wants to maintain accurate profile information. **Admin**: Monitors and ensures profile integrity.	
Precondition	User is logged in and has access to the profile page.	
Postcondition	User profile is updated.	
Main Success Scenario	**Actor Action**	**System Requirements**
	1. User updates their profile.	2. System saves the changes and reflects it on the profile.
Extensions	If invalid data is entered, the system prompts the user to correct it.	
Special Requirements	Data validation should be implemented to prevent incorrect entries.	

4. Get Personalized Content.

Usecase Name	Get Personalized Content	
Scope	User receives personalized content based on their past interactions and emotional state.	
Level	Primary	
Primary Actor	User	
Stakeholders and Interests	**User:** Receives content tailored to their needs and emotions. **System:** Provides relevant, helpful content.	
Precondition	User has a history of interactions that can be analyzed. But it not necessary.	
Postcondition	Personalized content is delivered.	
Main Success Scenario	**Actor Action**	**System Requirements**
	2. User sees the content.	1. The system analyzes user data and generates personalized recommendations.
Extensions	If the user has no history, default content is provided.	
Special Requirements	The recommendations must be accurate and fast.	

5. Get CBT Tools.

Usecase Name	Get CBT Tools	
Scope	User accesses cognitive behavioral therapy (CBT) tools within the app.	
Level	Primary	
Primary Actor	User	
Stakeholders and Interests	**User**: Uses CBT tools to improve mental health. **Admin**: Ensures tools are up-to-date and effective.	
Precondition	User is logged in and has accessed the CBT section.	
Postcondition	User successfully interacts with CBT tools.	
Main Success Scenario	**Actor Action**	**System Requirements**
	1. User selects and completes a CBT exercise.	
Extensions	If the user is in a crisis, the system directs them to crisis support.	
Special Requirements	Tools must be easy to navigate and scientifically backed.	

6. Report Chatbot Misbehavior

Usecase Name	Report Chatbot Misbehavior	
Scope	User reports inappropriate or incorrect behavior by the chatbot.	
Level	Primary	
Primary Actor	User	
Stakeholders and Interests	**User**: Ensures the chatbot behaves appropriately. **Admin**: Investigates and resolves chatbot issues.	
Precondition	User has interacted with the chatbot.	
Postcondition	Report is submitted and logged.	
Main Success Scenario	**Actor Action**	**System Requirements**
	1. User submits a report.	2. The system flags it for admin to review.

Extensions	If the report is urgent, the system flags it for immediate admin attention.
Special Requirements	Reporting tool should be easy and accessible

7. Users Journal.

Usecase Name	Journal of the user.	
Scope	User will be able to journal their daily life and mood.	
Level	primary	
Primary Actor	User	
Stakeholders and Interests	User: will journal their thoughts and mood.	
Precondition	User is logged in and goes to journaling interface.	
Postcondition	Journal is submitted and reflected.	
Main Success Scenario	**Actor Action**	**System Requirements**
	1. User inputs their thoughts and mood and other information the journal.	2. System submits and reflect it on the screen.
Extensions	If not submitted than notify it to the user	
Special Requirements	User interface must be user friendly and encouraging for user to put their thoughts.	

8. Manage User

Usecase Name	Manage Users	
Scope	Admin manages user accounts, including suspension or deletion.	
Level	Primary	
Primary Actor	Admin	
Stakeholders and Interests	**Admin**: Maintains user account integrity. **User**: Expects a safe and secure experience.	
Precondition	Admin has access to the user management system.	
Postcondition	User accounts are updated or removed.	
Main Success Scenario	**Actor Action**	**System Requirements**
	1. Admin suspends, deletes, or updates user accounts as necessary.	

Extensions	System logs changes for security purposes.
Special Requirements	Data privacy and security protocols must be in place.

9. Moderate Users.

Usecase Name	Moderate Users	
Scope	Expert moderates user interactions and intervenes when necessary.	
Level	Primary	
Primary Actor	Expert	
Stakeholders and Interests	**Expert**: Ensures conversations are safe and productive. **User**: Expects empathetic and supportive interactions.	
Precondition	Expert has access to user conversations.	
Postcondition	Conversations are moderated.	
Main Success Scenario	**Actor Action**	**System Requirements**
	2. Expert takes action accordingly.	1. System shows potential vulnerable chats.
Extensions	If the situation is severe, the expert escalates to crisis support.	
Special Requirements	Access controls must ensure experts can only view relevant data.	

10. Answers Queries.

Usecase Name	Answer Queries/Feedback	
Scope	Expert answers queries or feedback from users.	
Level	Primary	
Primary Actor	Expert	
Stakeholders and Interests	**User**: Receives accurate and helpful responses. **Expert**: Provides insight and resolves issues.	
Precondition	User have submitted a query.	
Postcondition	Query is resolved and user is satisfied.	
	Actor Action	**System Requirements**

Main Success Scenario	1. Expert Responds to the query.	2. System submits the query.
Extensions	If the query is outside the expert's scope, it is escalated.	
Special Requirements	Expert must have access to relevant tools and information.	

11. Evaluate the need for Crisis Support.

Usecase Name	Evaluate the Need for Support	
Scope	System evaluates if a user requires crisis support.	
Level	Primary	
Primary Actor	System	
Stakeholders and Interests	**User**: Receives appropriate crisis support. **System**: Provides timely interventions.	
Precondition	User is in conversation with the chatbot, and emotional distress is detected.	
Postcondition	User is directed to crisis support if needed.	
Main Success Scenario	**Actor Action**	**System Requirements**
	1. User has activities in journaling or chatbot.	2. System detects distress and offers crisis support.
Extensions	If the user is not in immediate crisis, supportive content is provided.	
Special Requirements	The system must have accurate emotional detection models.	

12. Backup Conversations.

Usecase Name	Backup Conversations	
Scope	System backs up conversations for future retrieval and security.	
Level	Primary	
Primary Actor	System	
Stakeholders and Interests	**Admin**: Ensures conversations are securely backed up. **User**: Expects data to be securely stored.	
Precondition	Conversation has occurred.	
Postcondition	Conversations have backed up and stored.	
	Actor Action	**System Requirements**

Main Success Scenario	1. User has a convo with the chatbot.	2. System automatically backs up conversation data.
Extensions	If the backup fails, the system retries or alerts the admin.	
Special Requirements	Backups must be encrypted and securely stored.	

13. Response based on emotion.

Usecase Name	Response based on Emotion	
Scope	Chatbot responds empathetically based on detected emotions.	
Level	Primary	
Primary Actor	Chatbot	
Stakeholders and Interests	**User**: Expects empathetic and relevant responses. **System**: Ensures emotional accuracy in responses.	
Precondition	User has initiated a conversation.	
Postcondition	Chatbot responds empathetically.	
Main Success Scenario	**Actor Action**	**System Requirements**
	1. User starts a conversation,	2. System determined the emotion of the convo. 3. Responses in the accurate empathetic manner rather in voice or text depending on the nature of the response from the user.
Extensions	If emotion is not detected, chatbot provides a neutral response.	
Special Requirements	Emotion detection module should be accurate.	

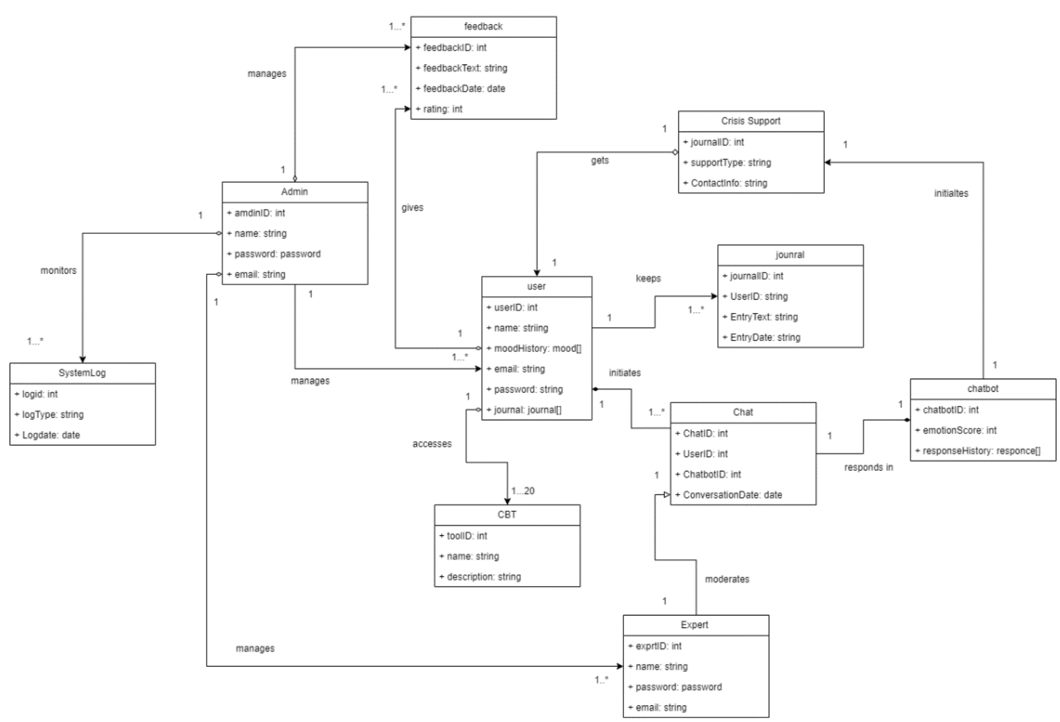
14. Manage Feedback

Usecase Name	Manage Feedback
Scope	Admin reviews and maanges user feedback.
Level	Primary
Primary Actor	Admin
Stakeholders and Interests	**Admin**: Needs to ensure feedback is appropriately addressed.

	User: Wants their feedback considered.	
Precondition	Feedback has been submitted by users.	
Postcondition	Feedback is reviewed and acted upon.	
Main Success Scenario	**Actor Action**	**System Requirements**
	1. Admin views and resolves feedback issues.	
Extensions	Feedback requiring expert attention is escalated.	
Special Requirements	Feedback management tools must be intuitive for admins.	

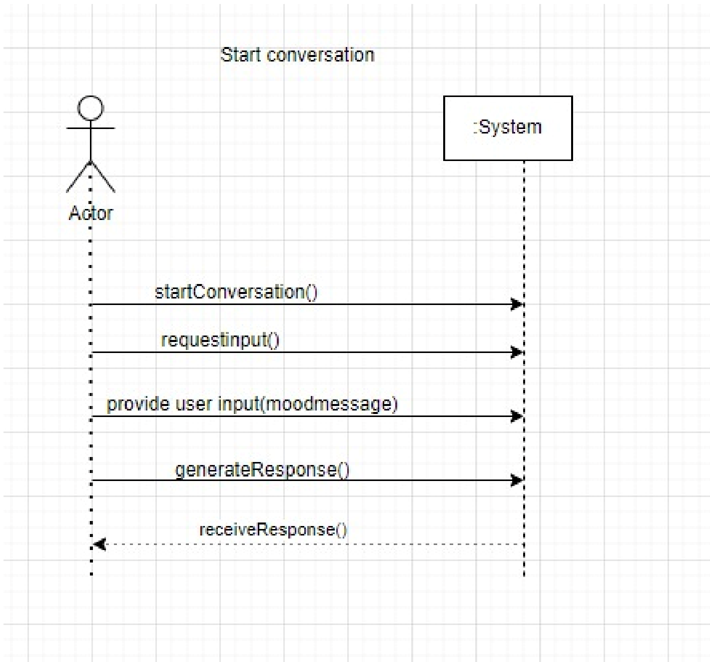
Domain Model:

Domain Model

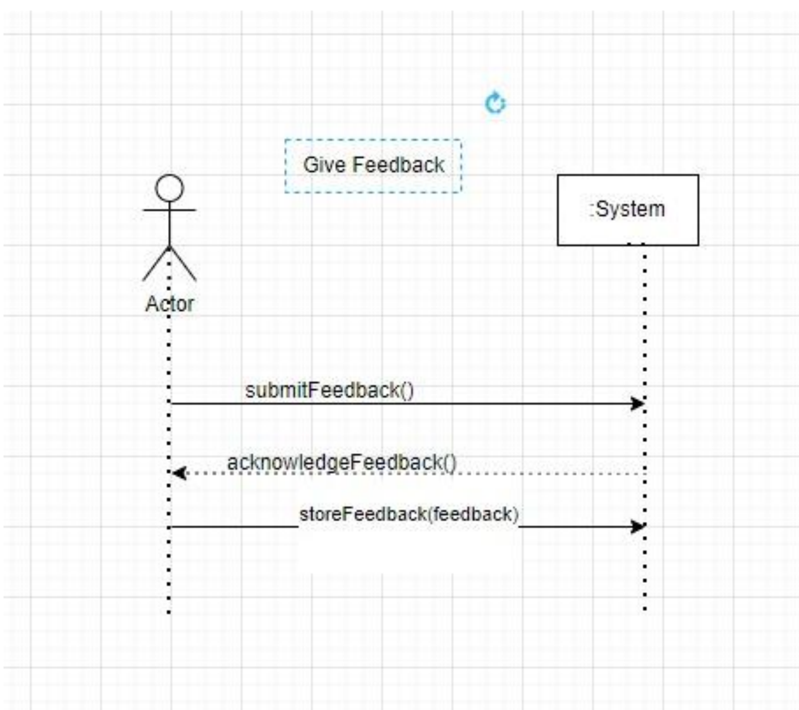


System Sequence Diagrams (SSD):

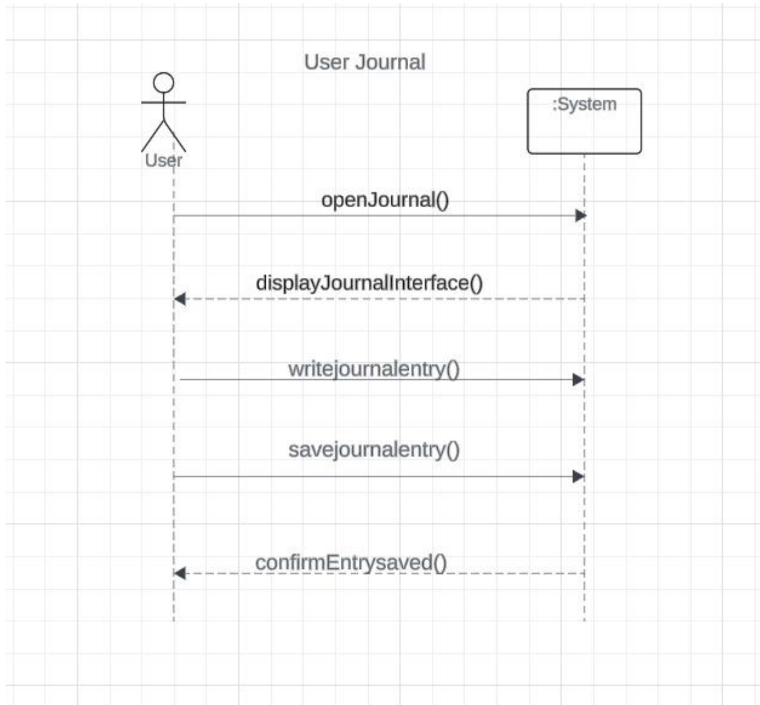
1. Start Conversation



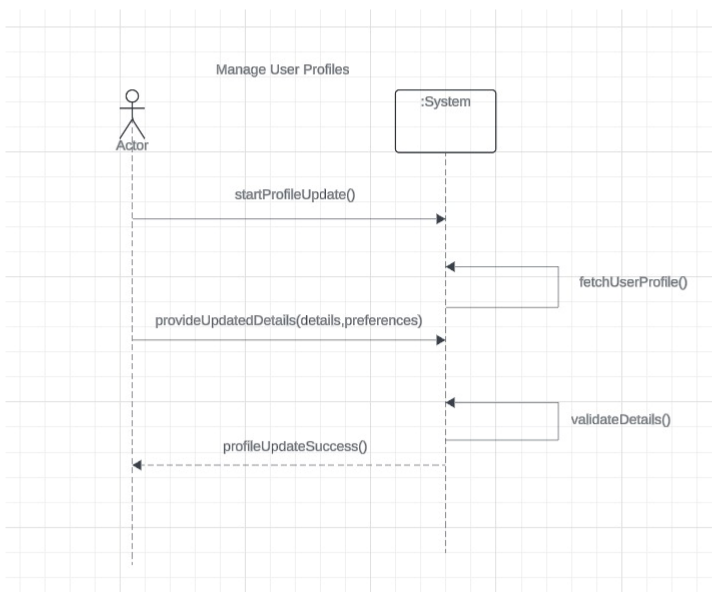
2. Give Feedback



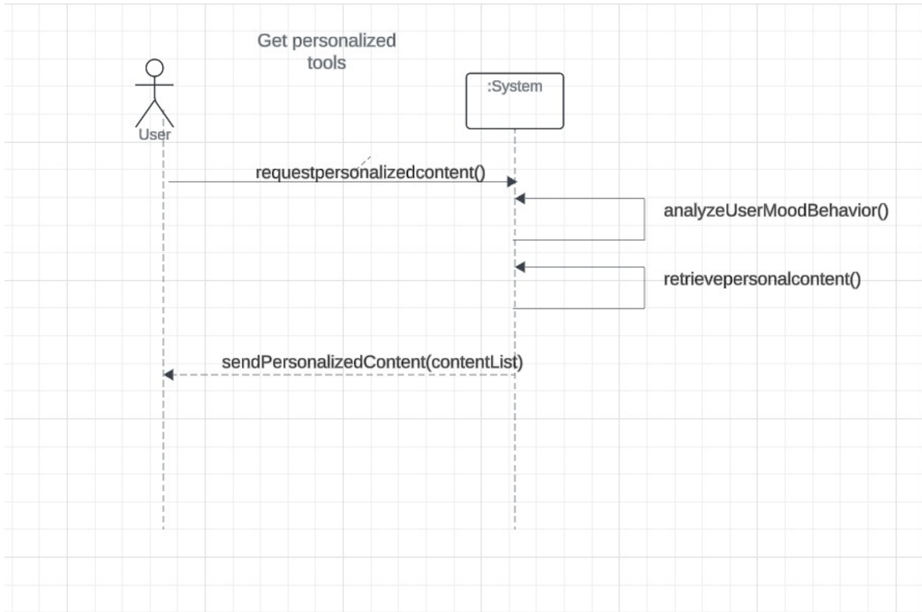
3. User Journal



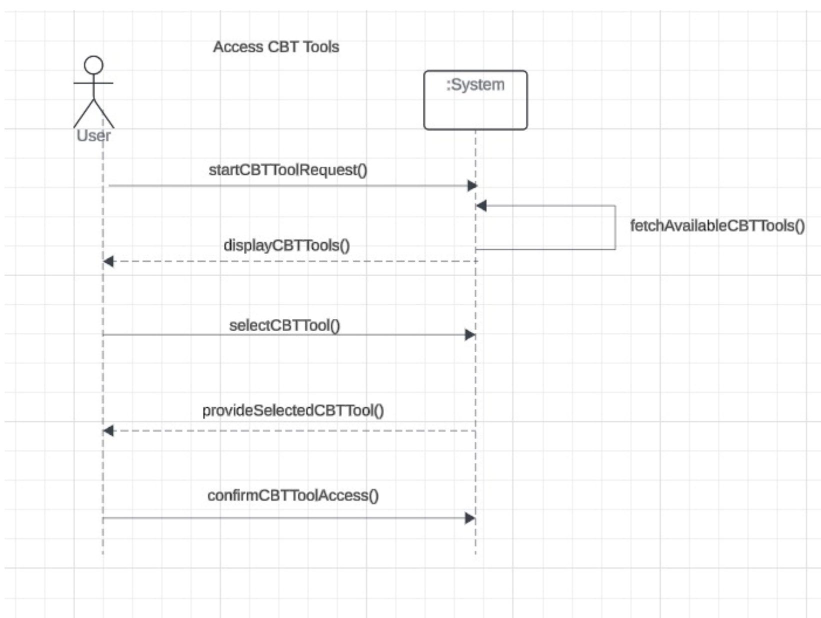
4. Manage User Profiles



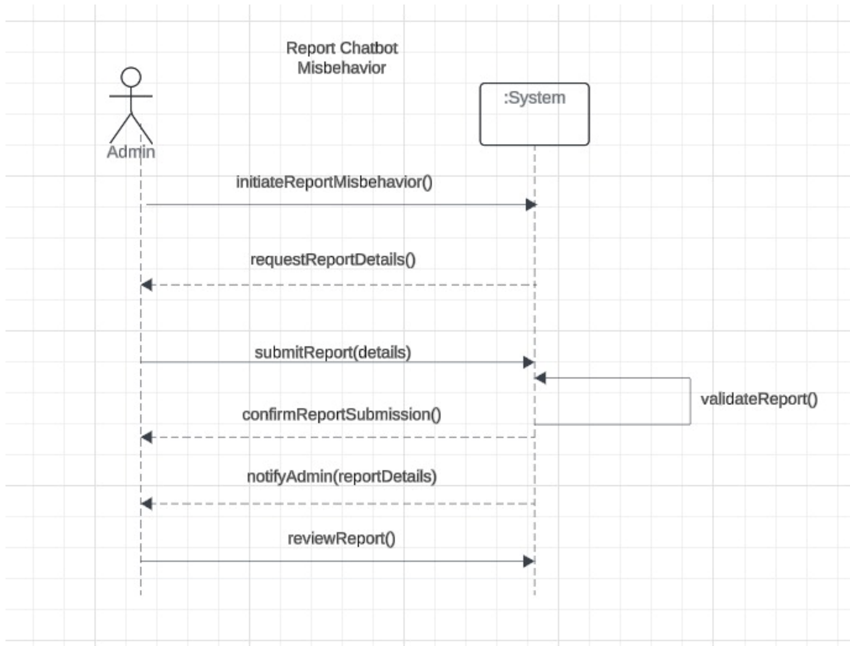
5. Get Personalized Content



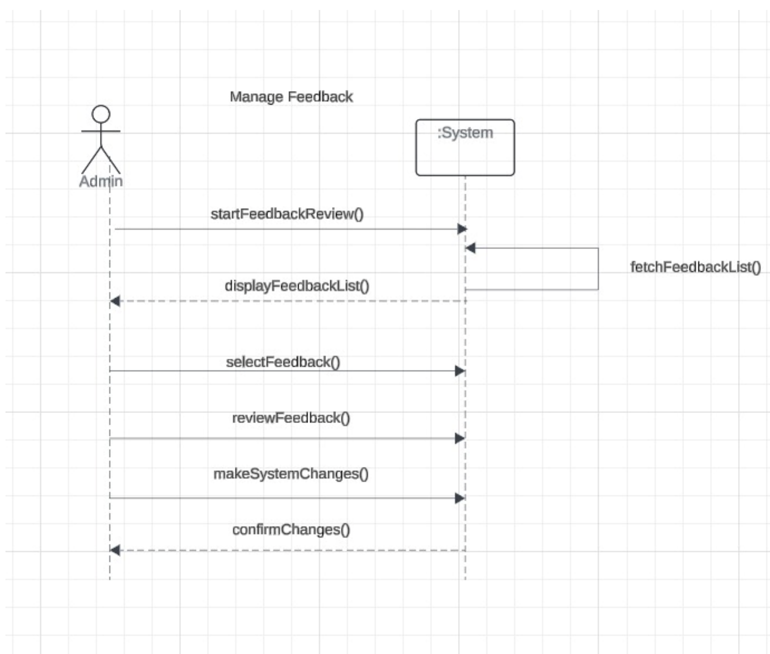
6. Access CBT Tools



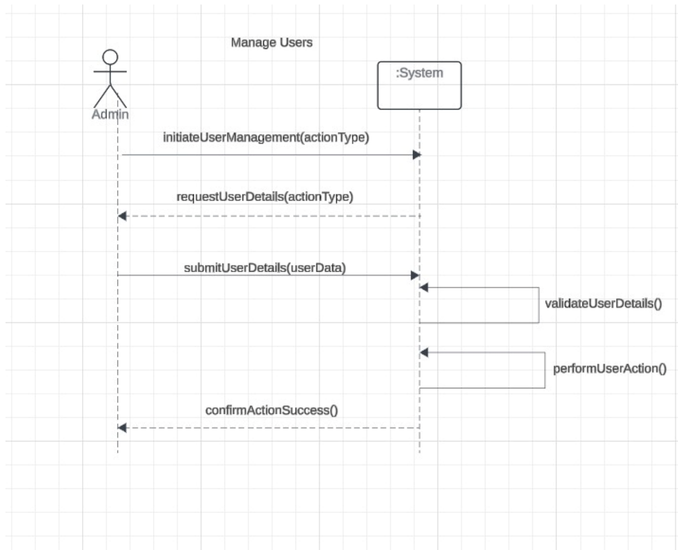
7. Report Chatbot Misbehavior



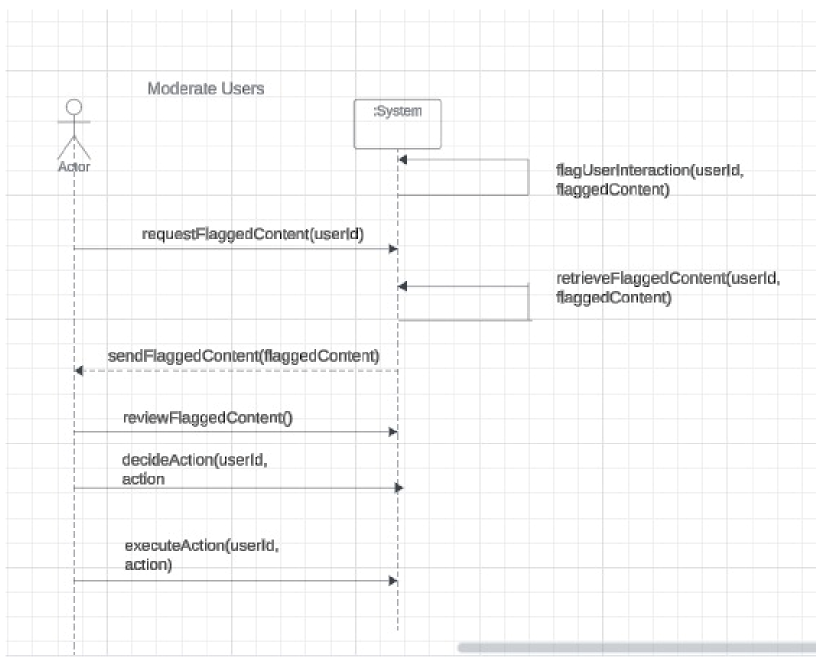
8. Manage Feedback



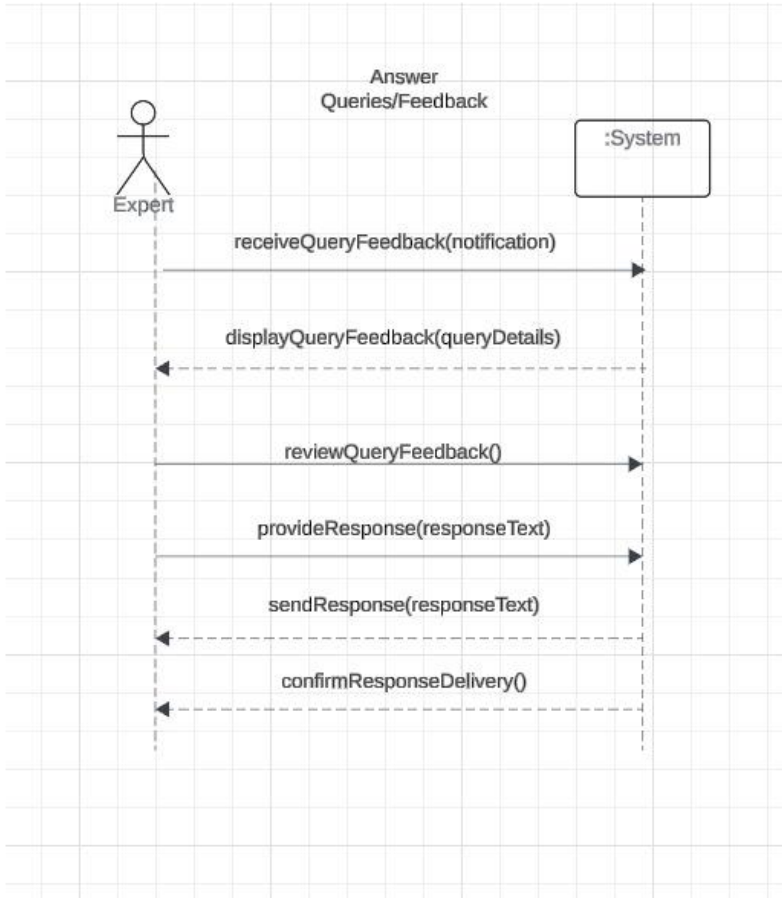
9. Manage Users



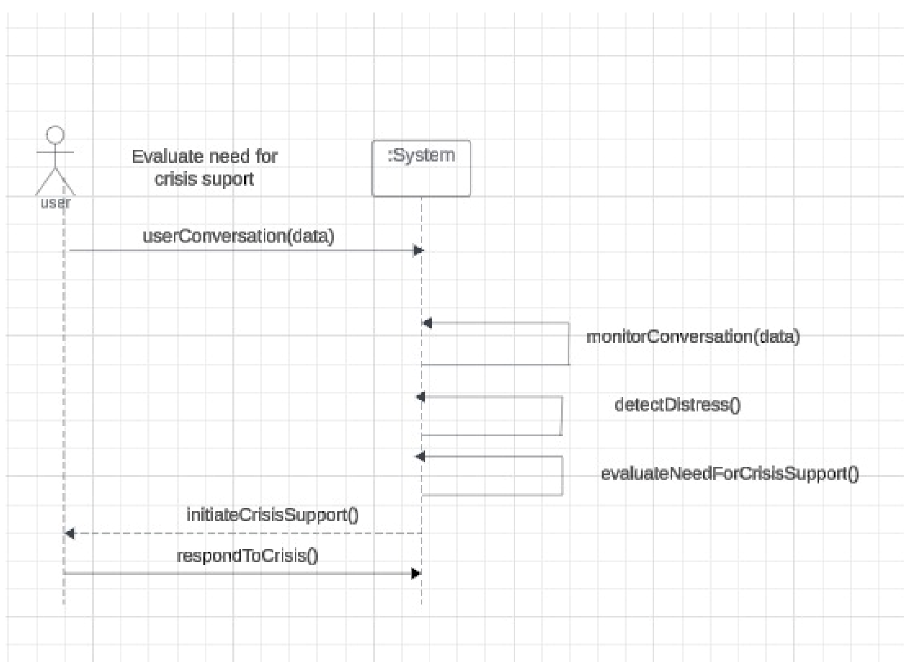
10. Moderate Users



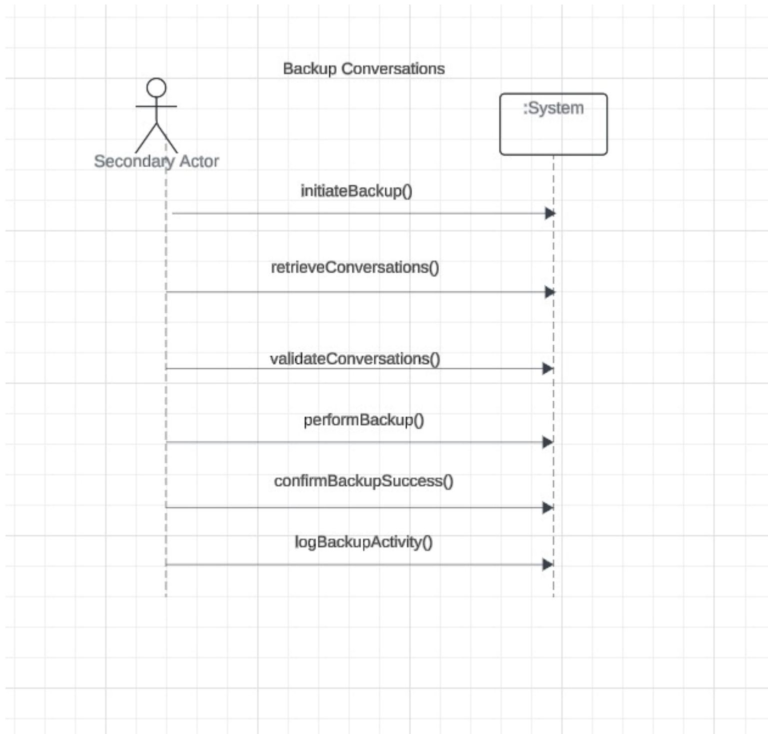
11. Answer Queries/Feedback



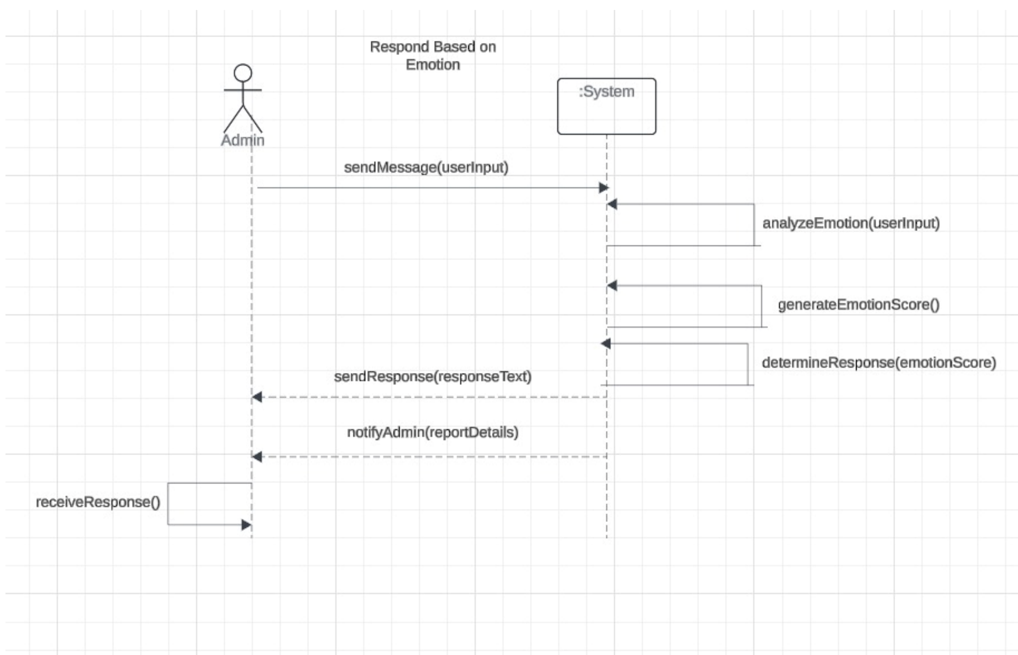
12. Evaluate Need for Crisis Support



13. Backup Conversations

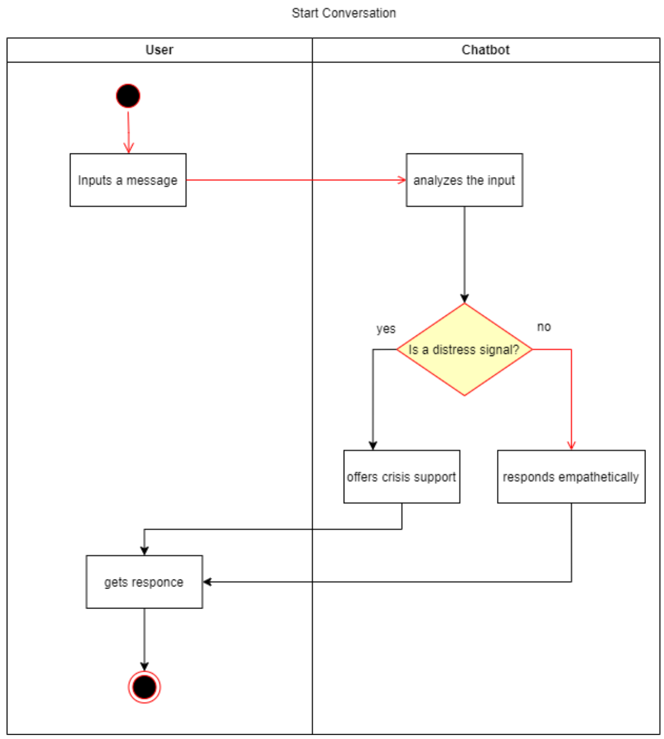


14. Respond Based on Emotion

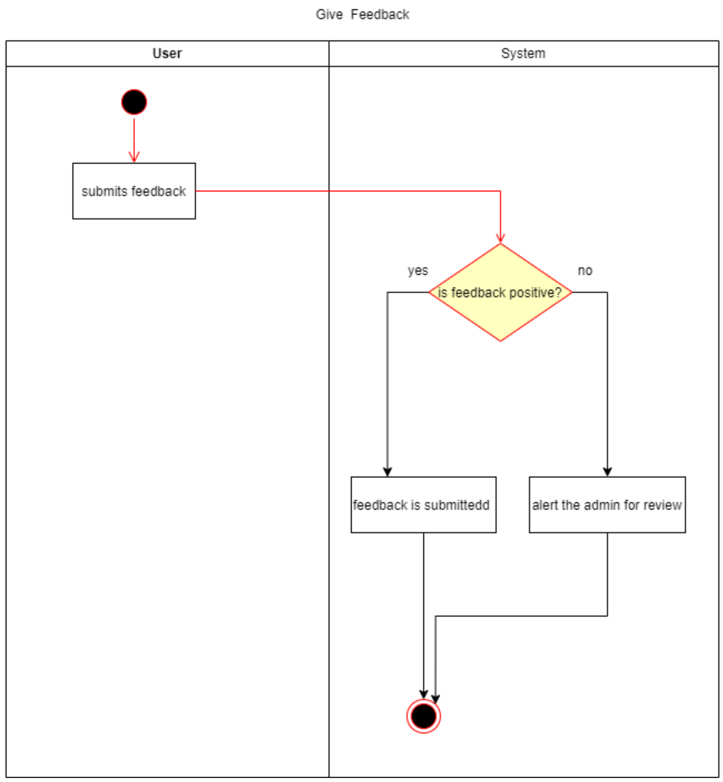


Activity / Data Flow Diagram:

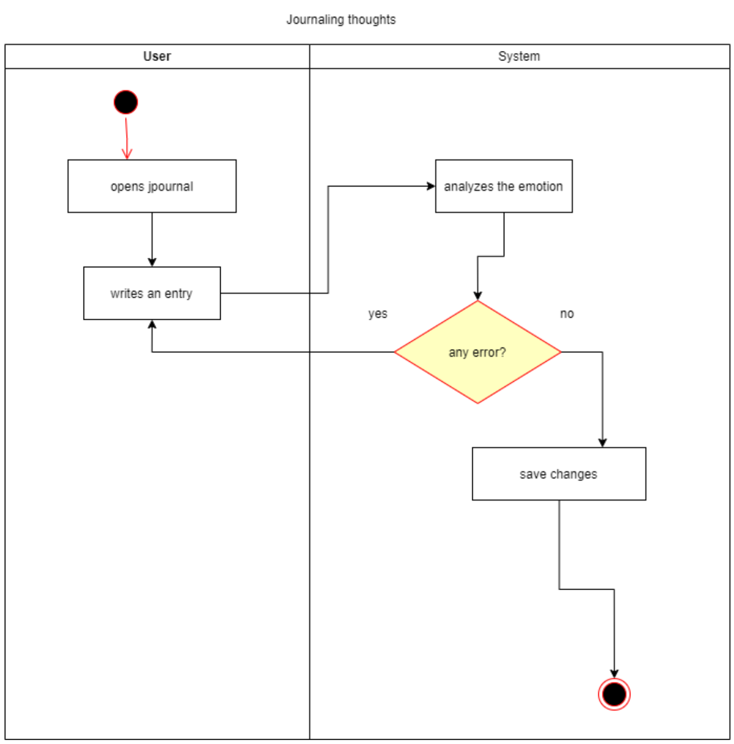
1. Start Conversation



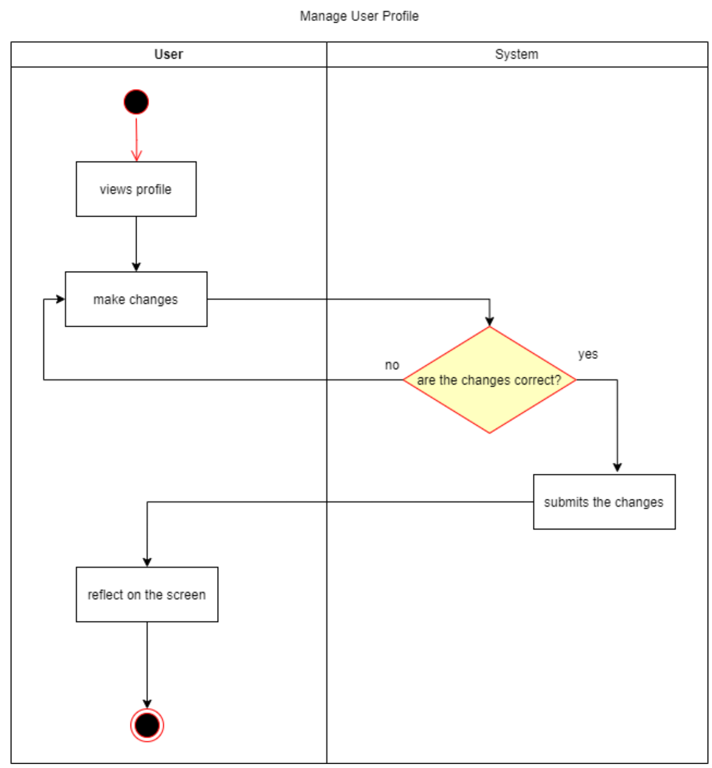
2. Give Feedback



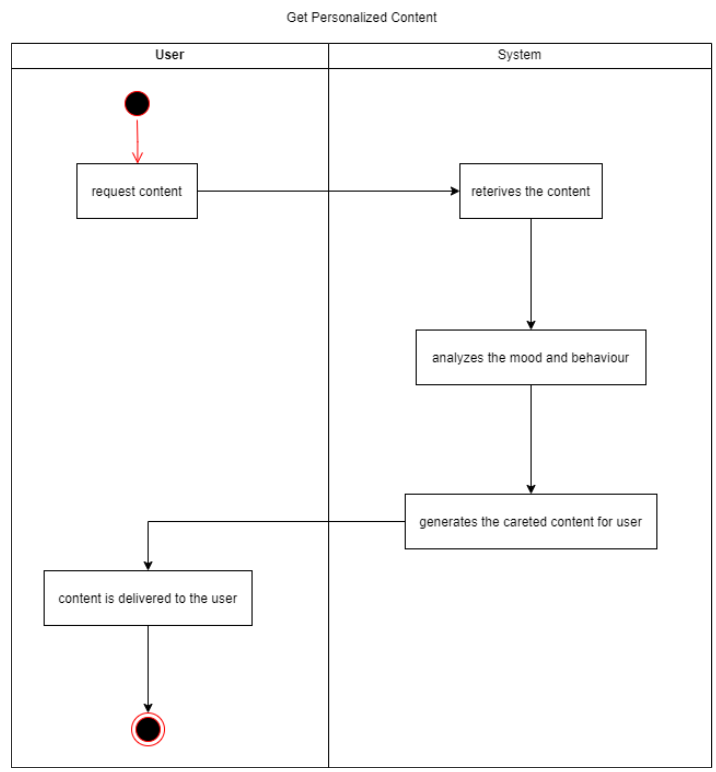
3. User Journal



4. Manage User Profiles



5. Get Personalized Content



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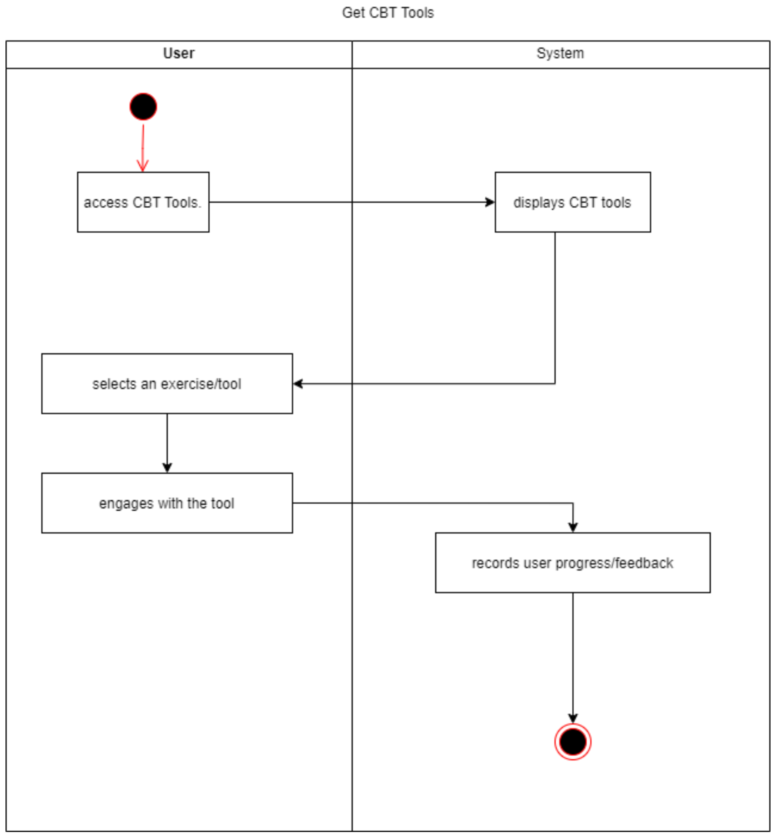
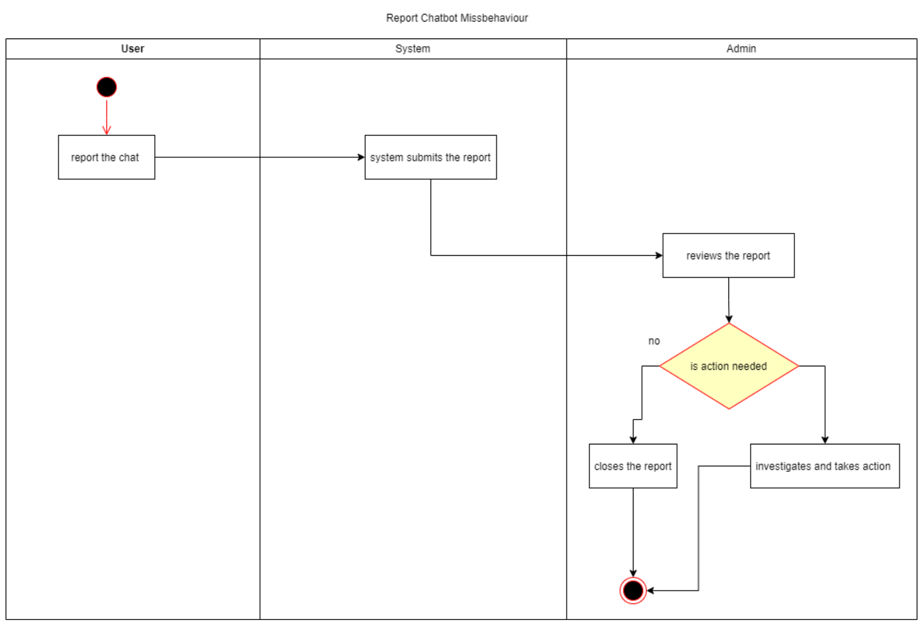
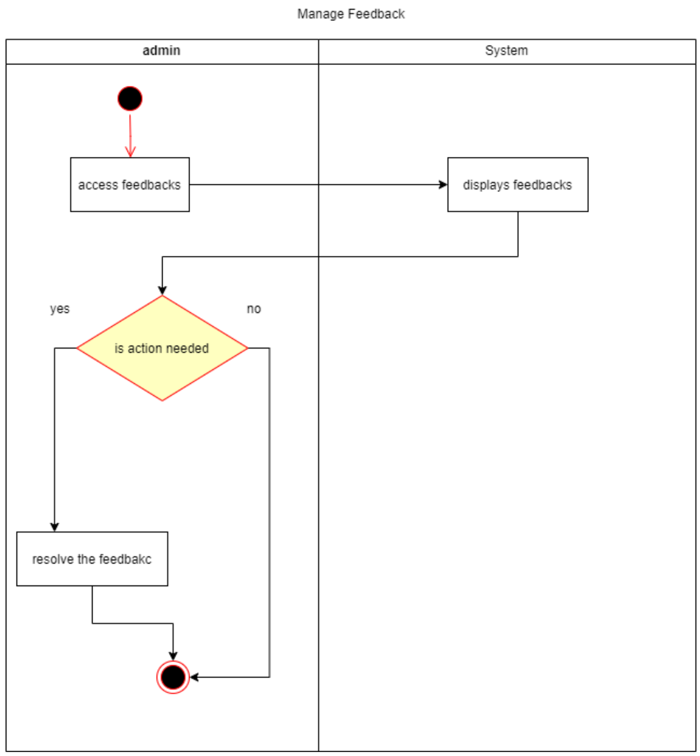


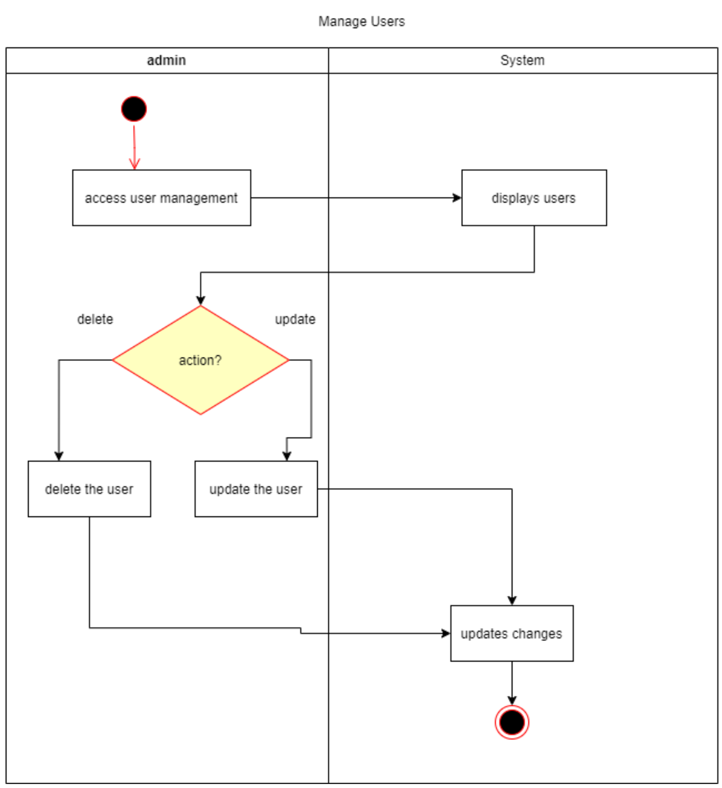
7. Report Chatbot Misbehavior



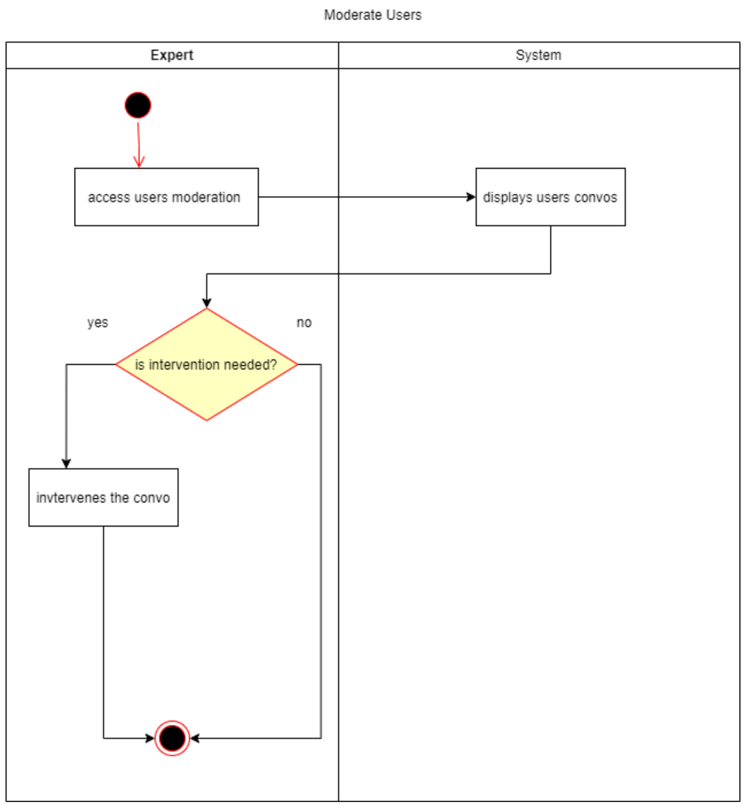
8. Manage Feedback



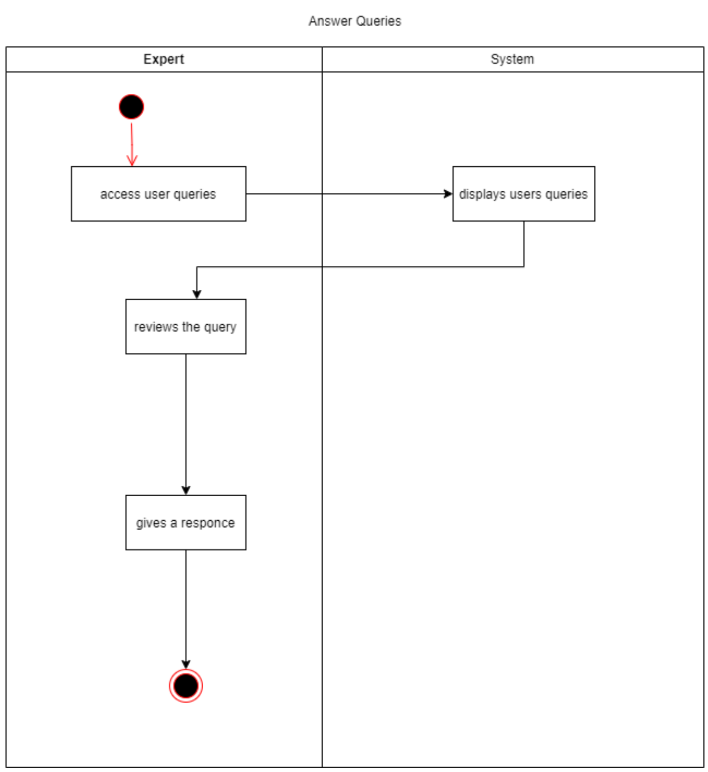
9. Manage Users



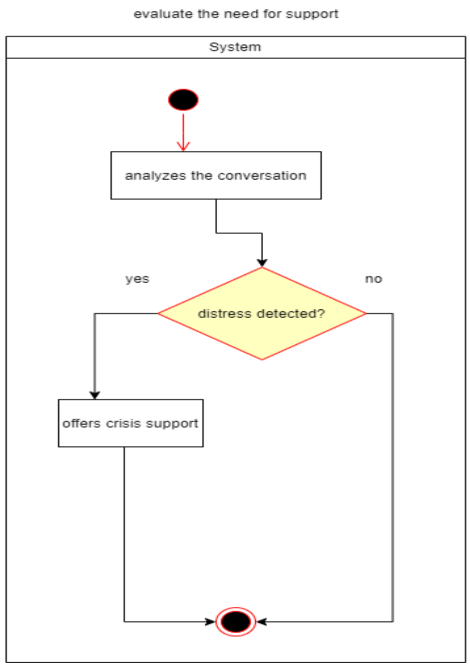
10. Moderate Users



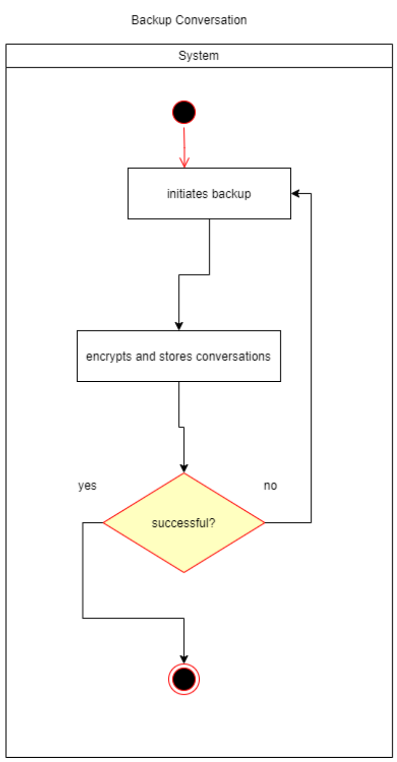
11. Answer Queries/Feedback



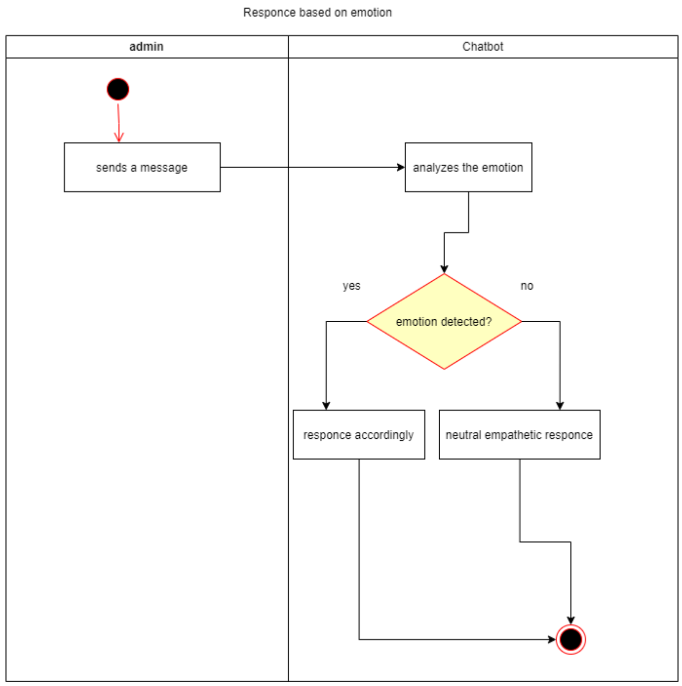
12. Evaluate Need for Crisis Support



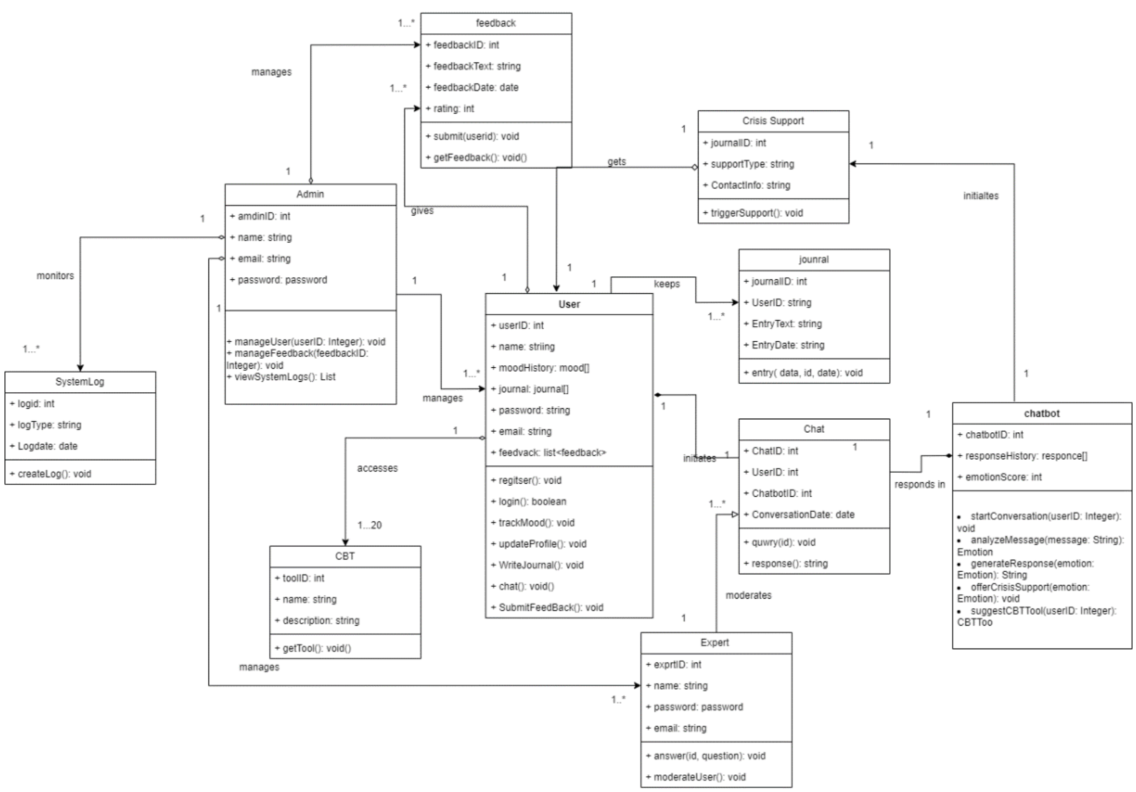
13. Backup Conversations



14. Respond Based on Emotion

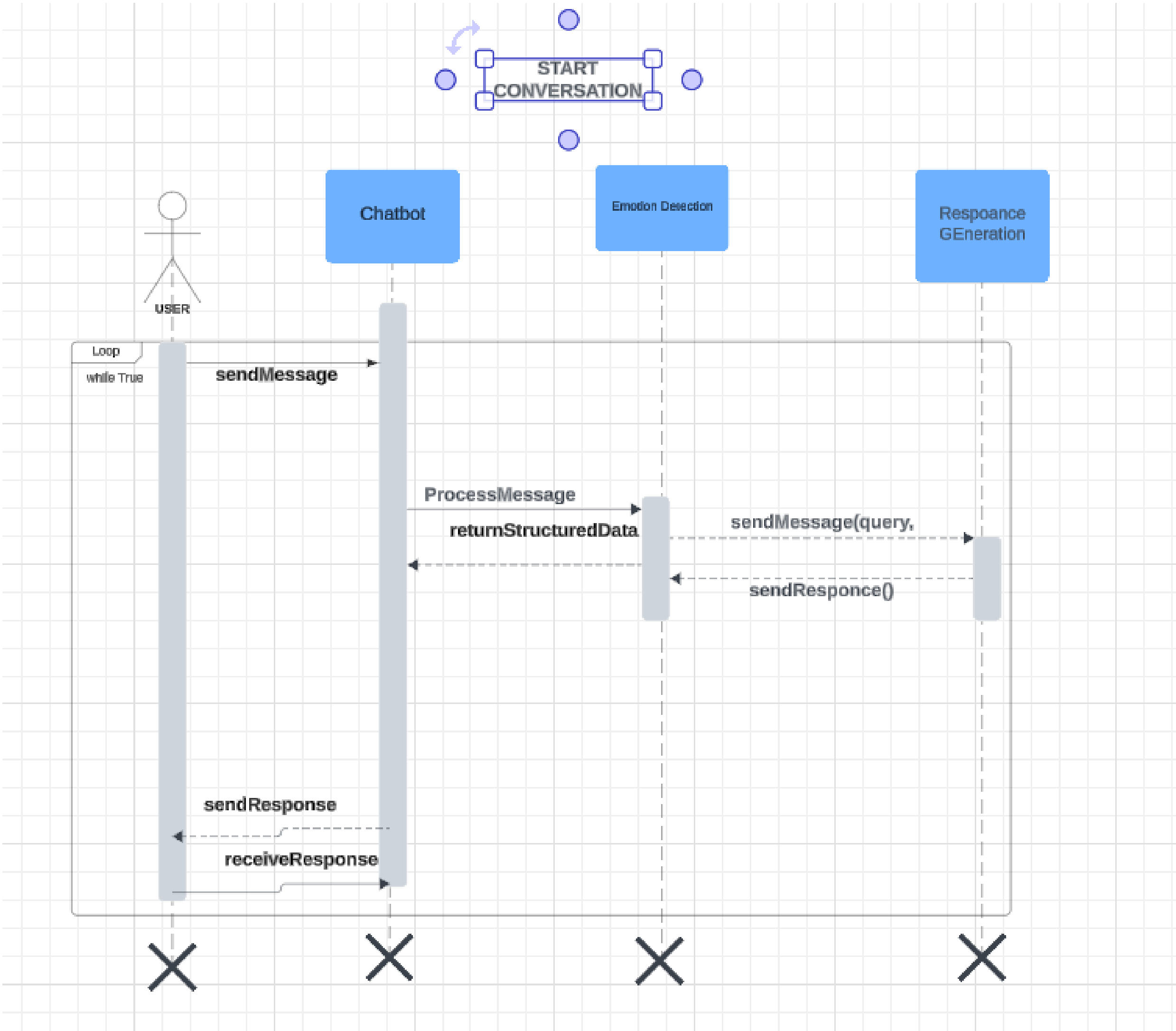


Class Diagram:

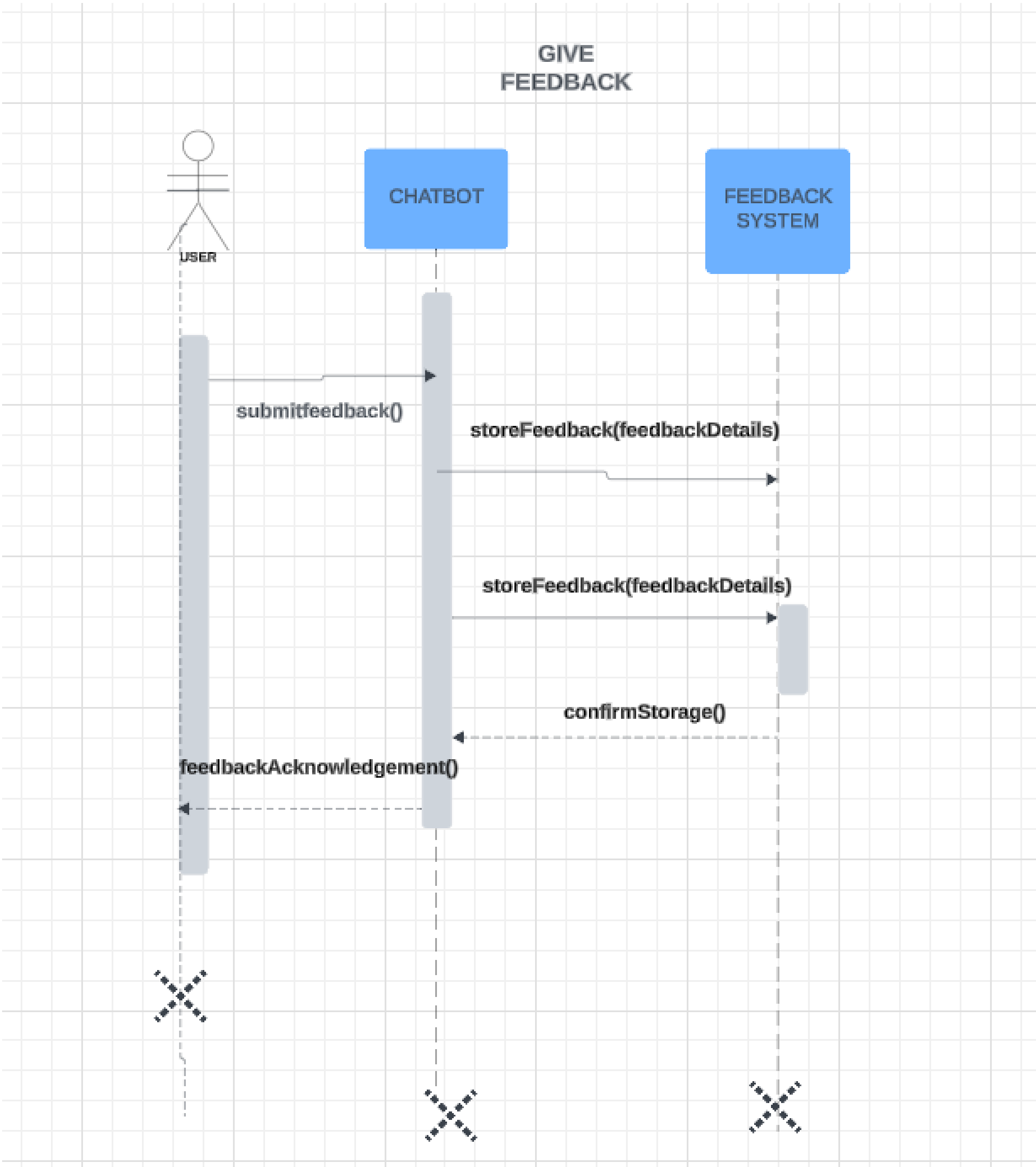


System Diagram (SD):

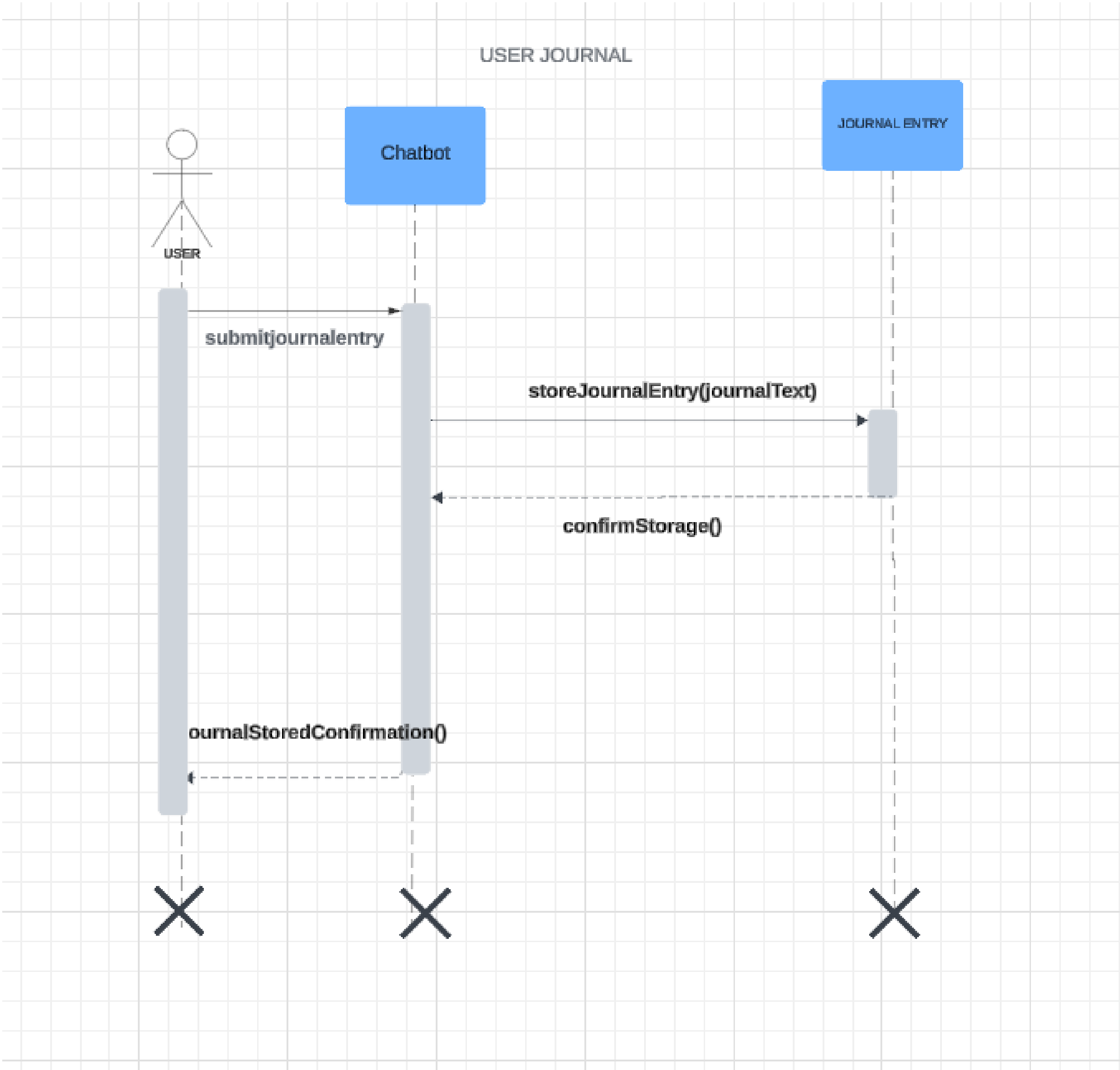
1. Start Conversation



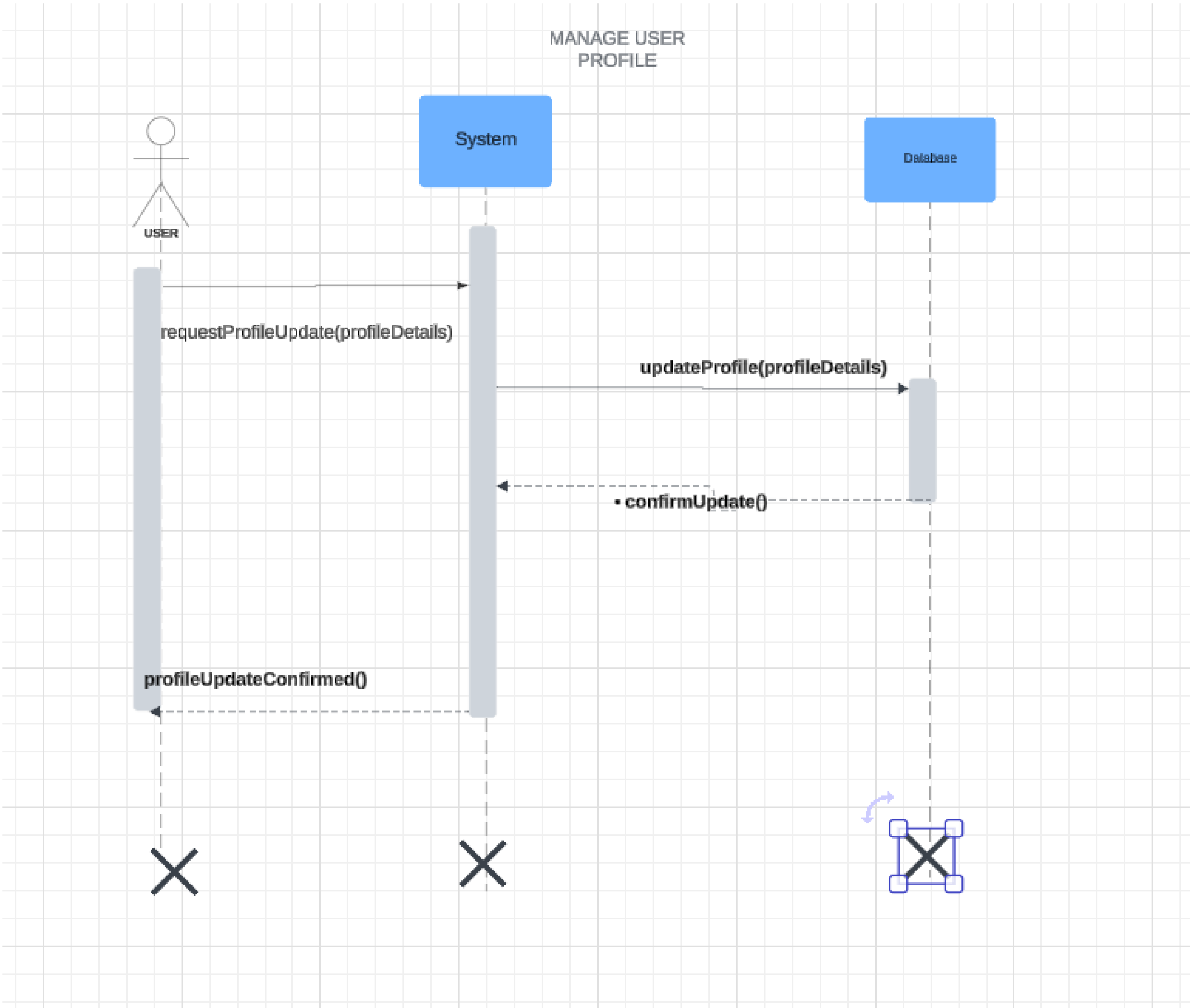
2. Give Feedback



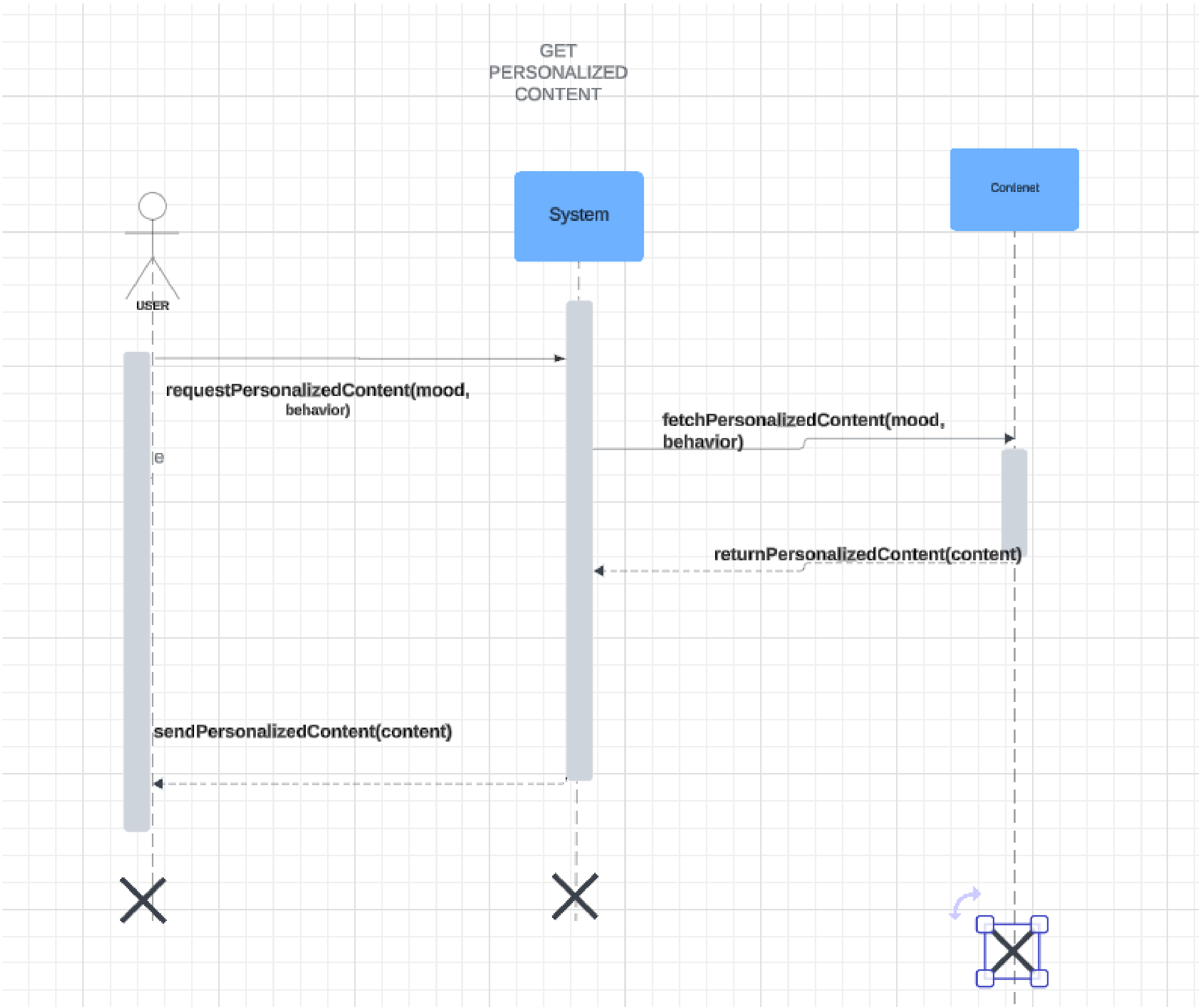
3. User Journal



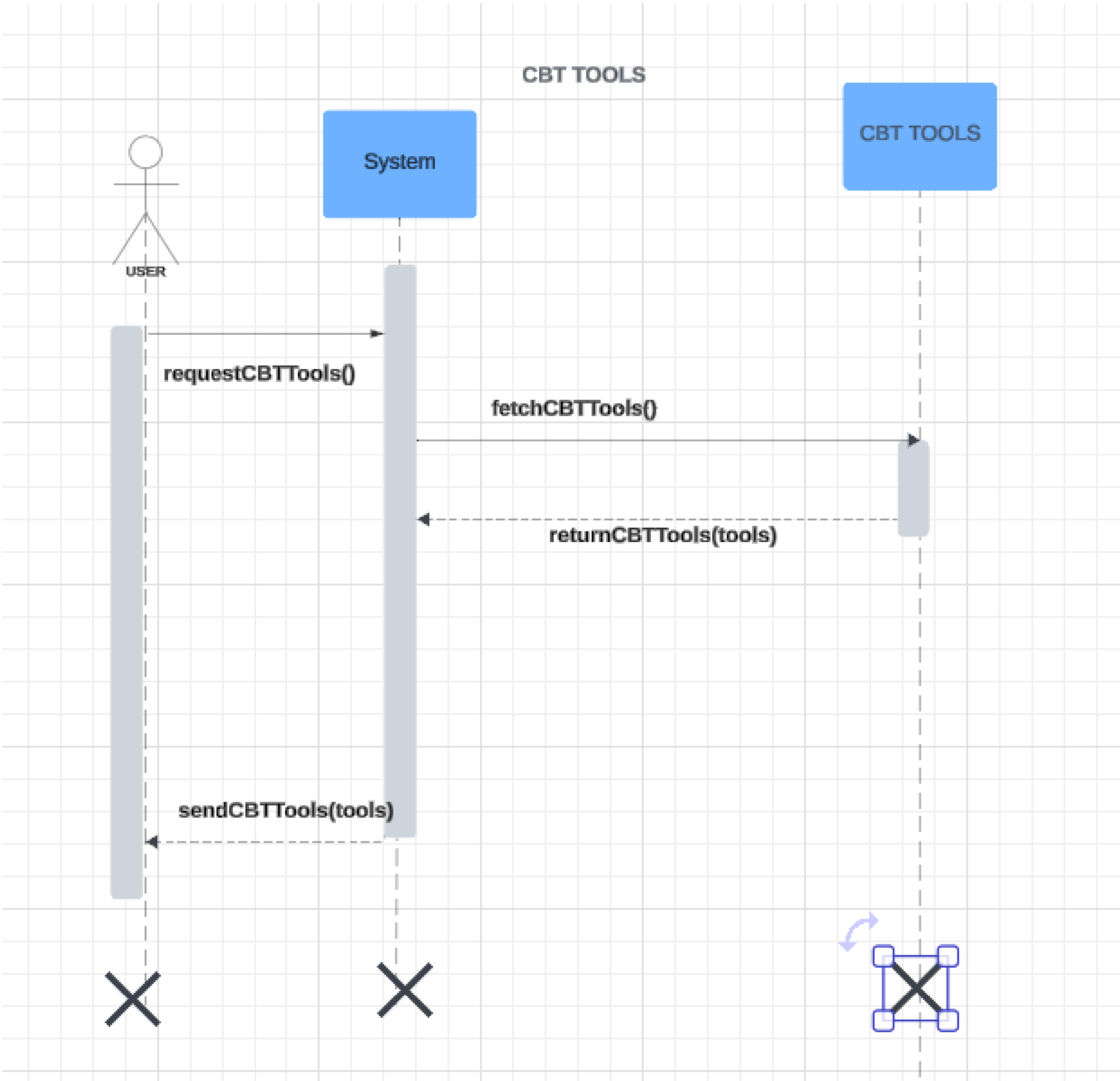
4. Manage User Profiles



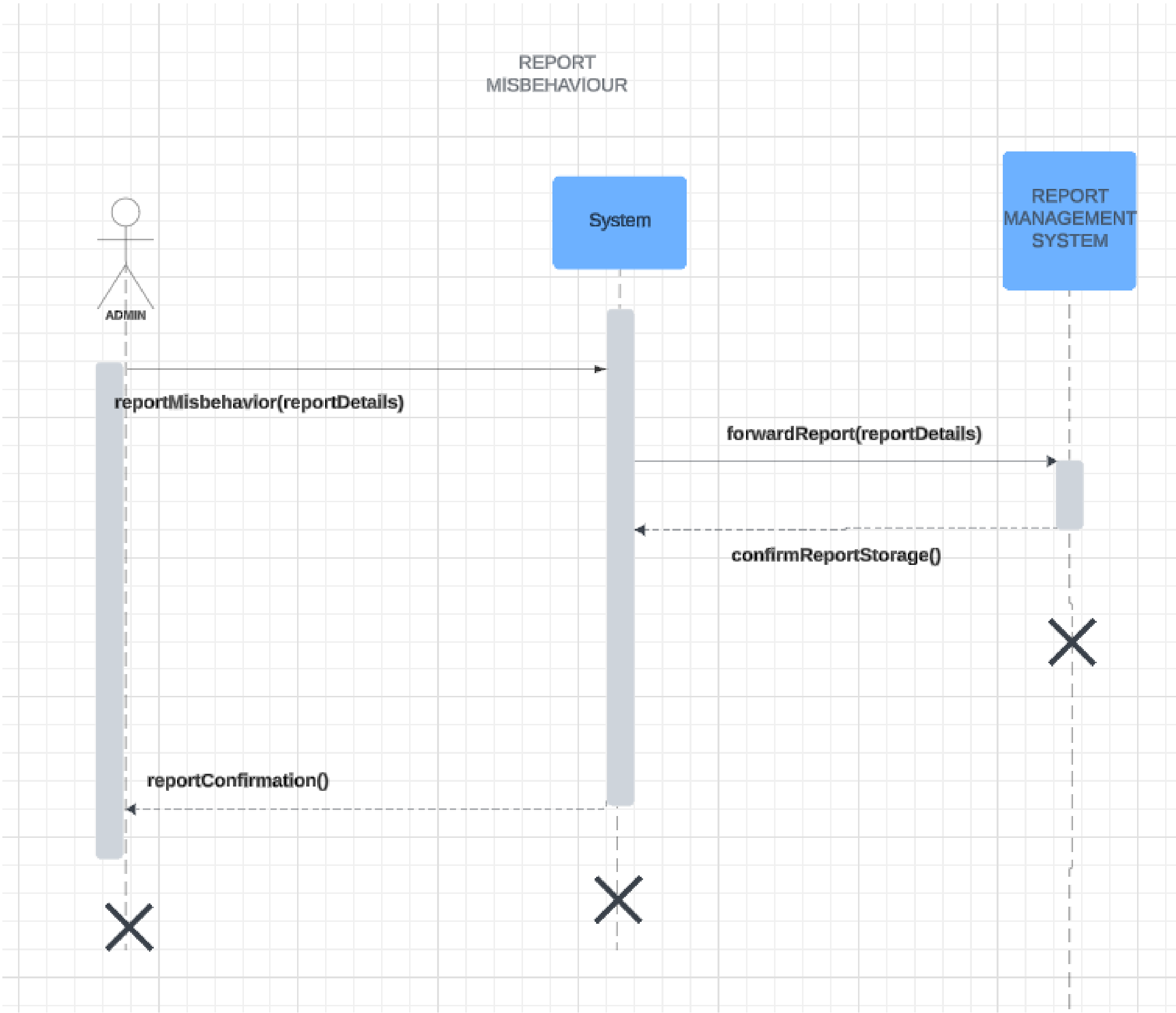
5. Get Personalized Content



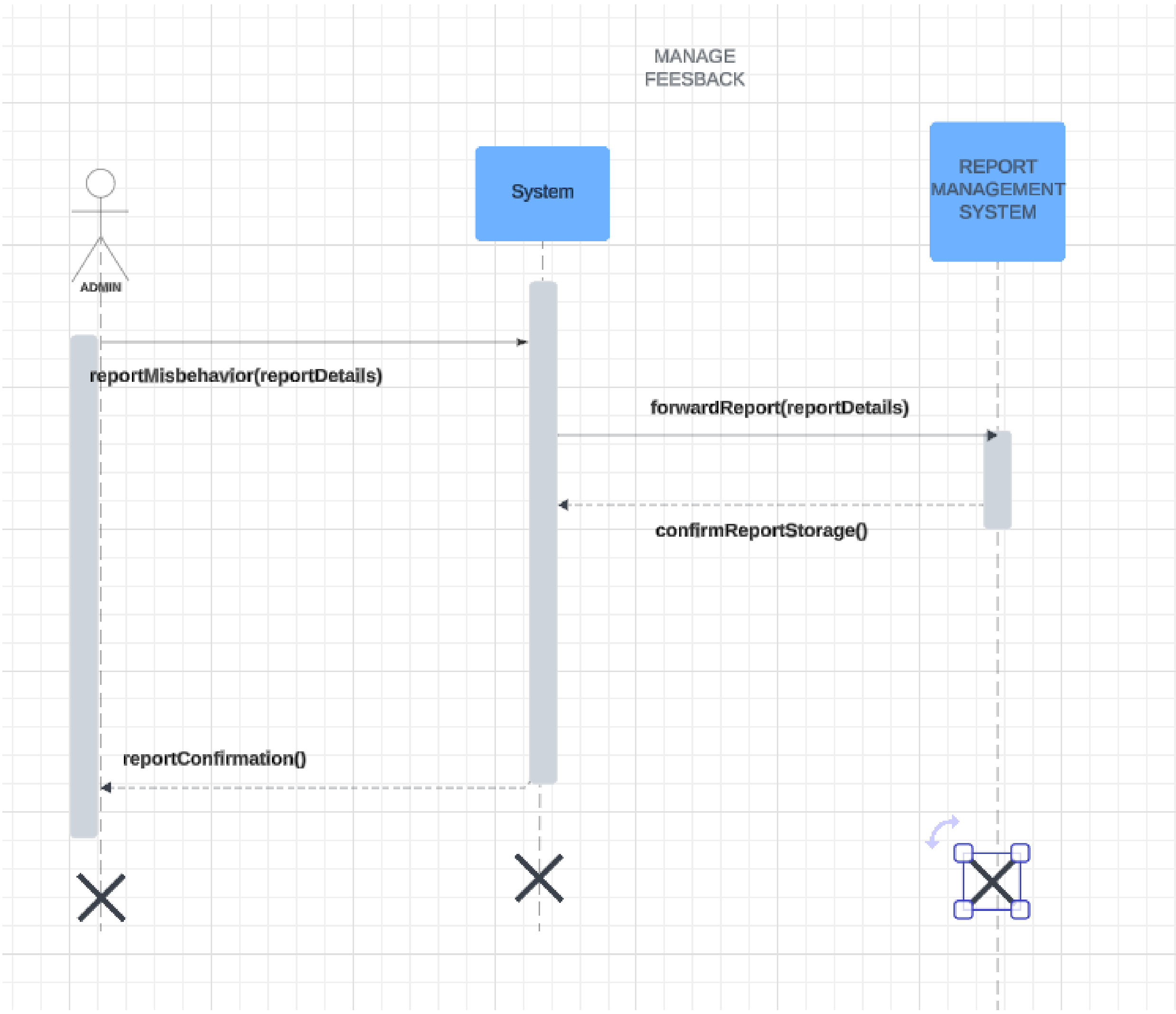
6. Access CBT Tools



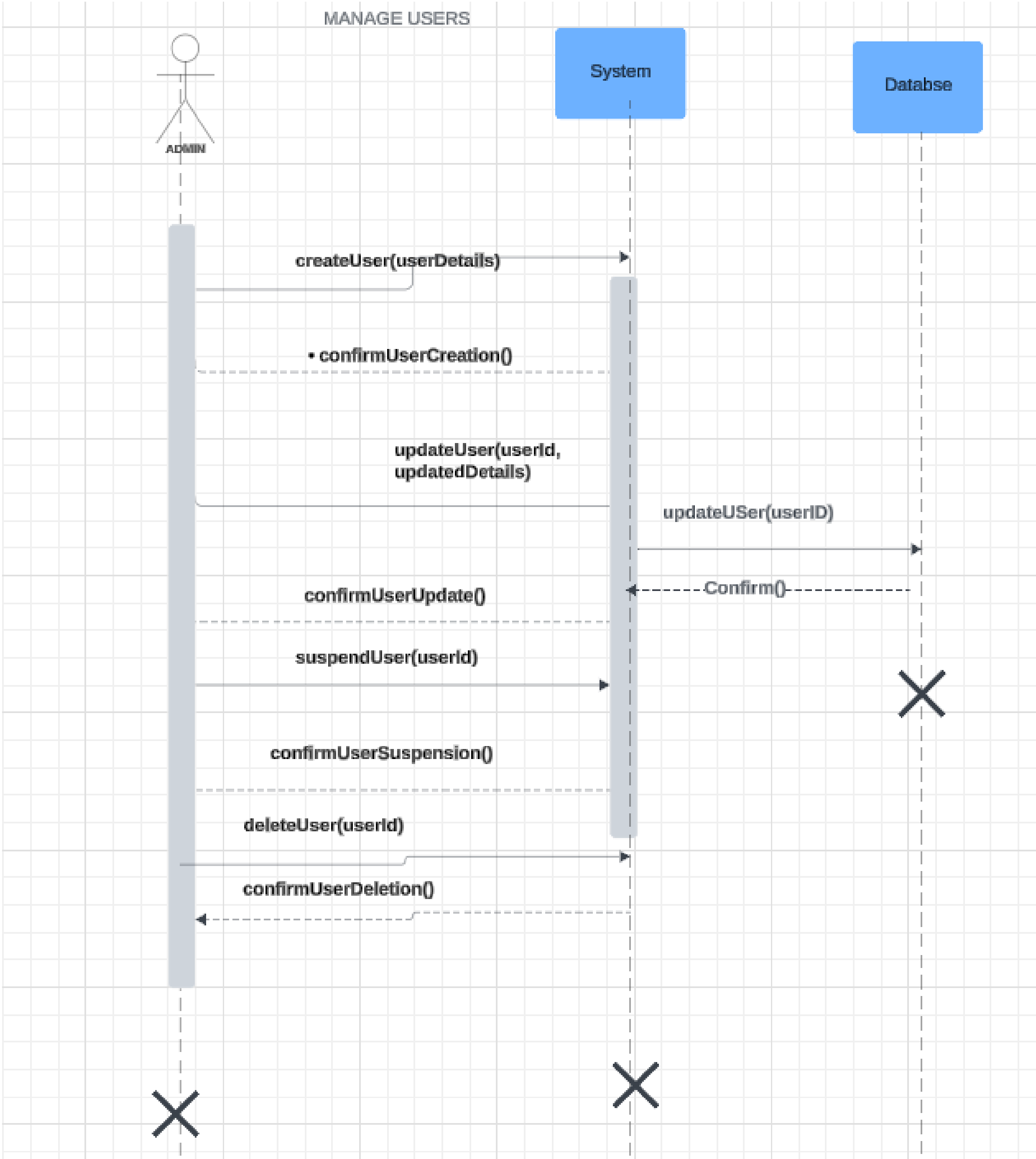
7. Report Chatbot Misbehavior



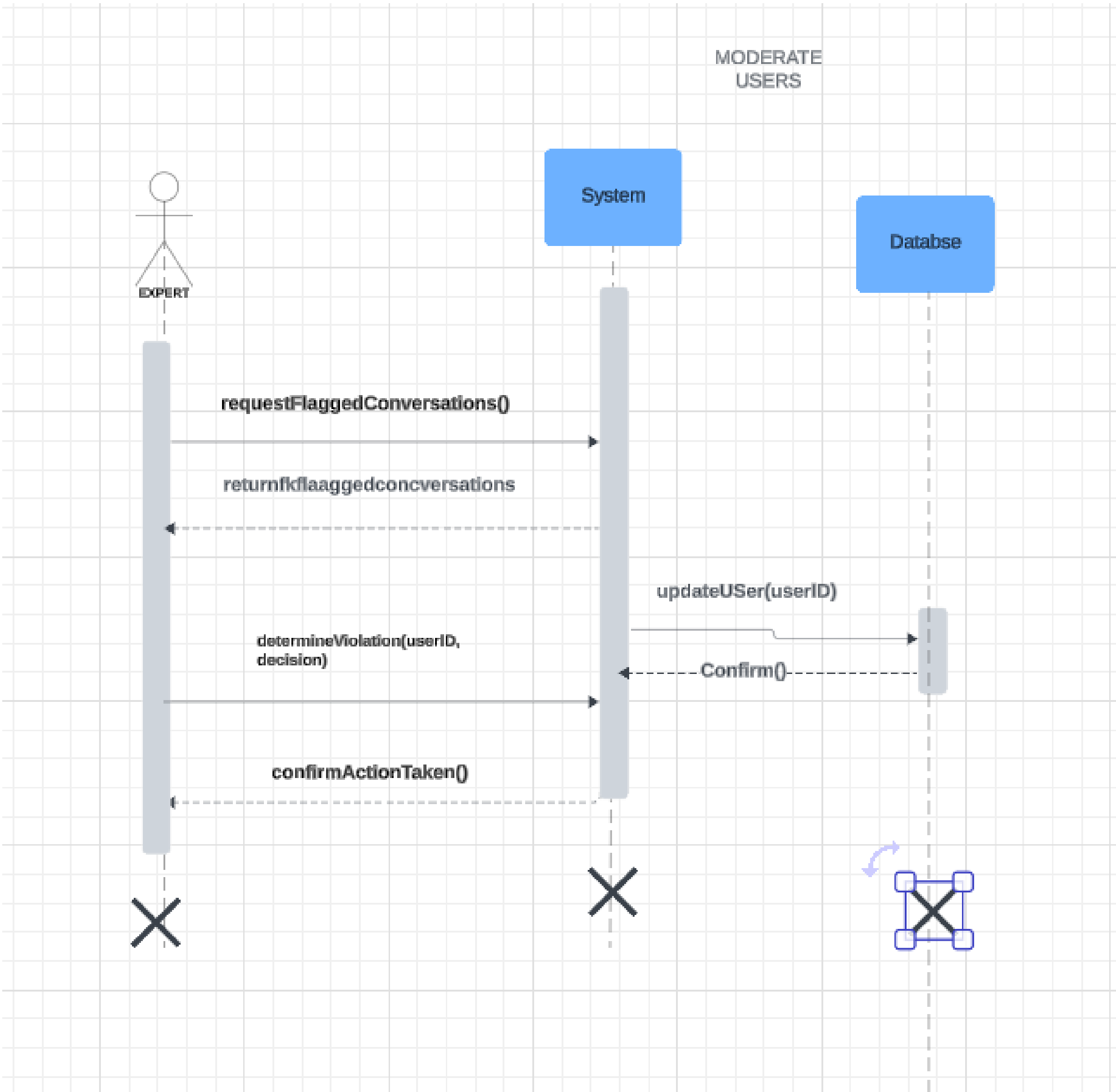
8. Manage Feedback



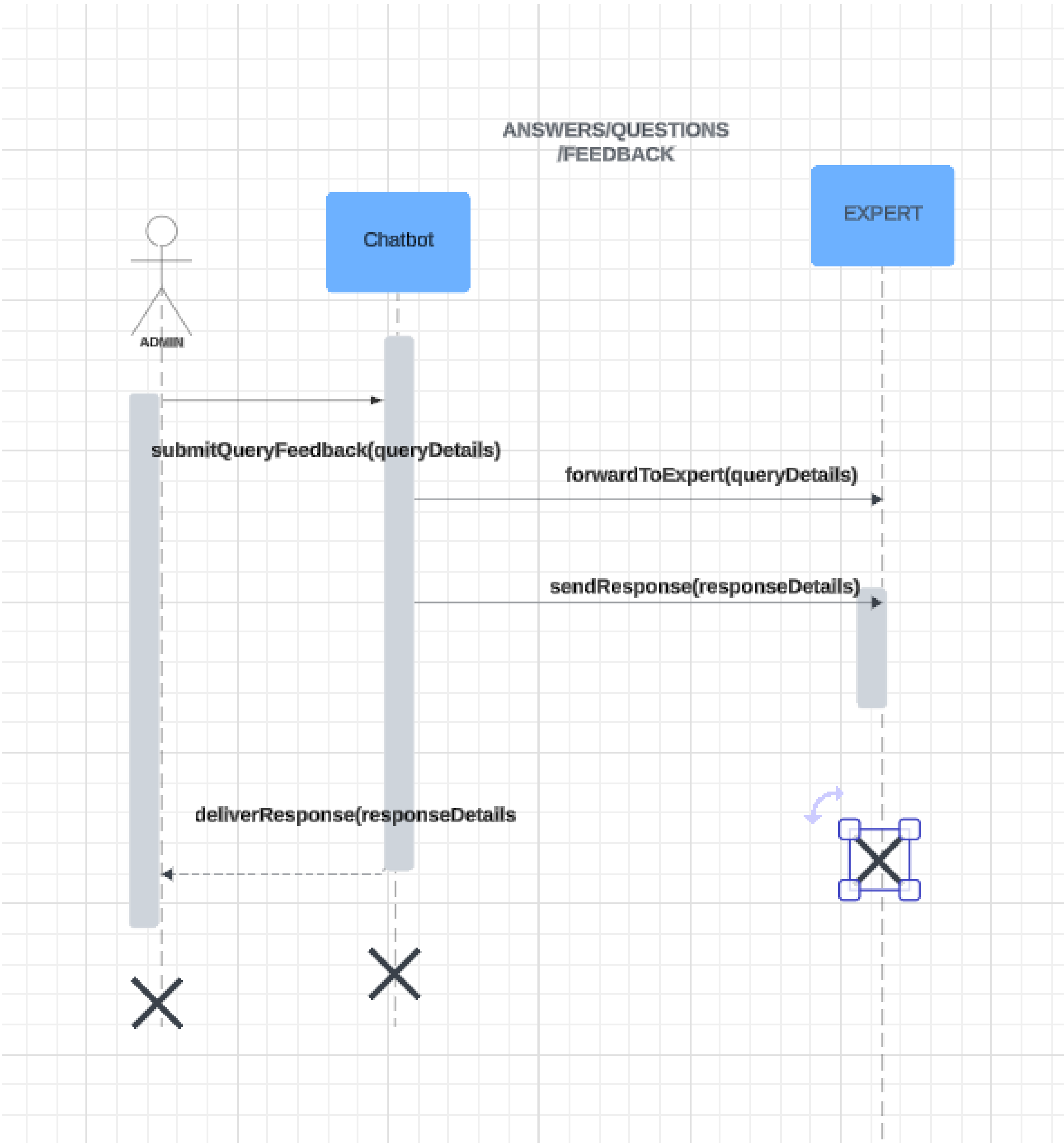
9. Manage Users



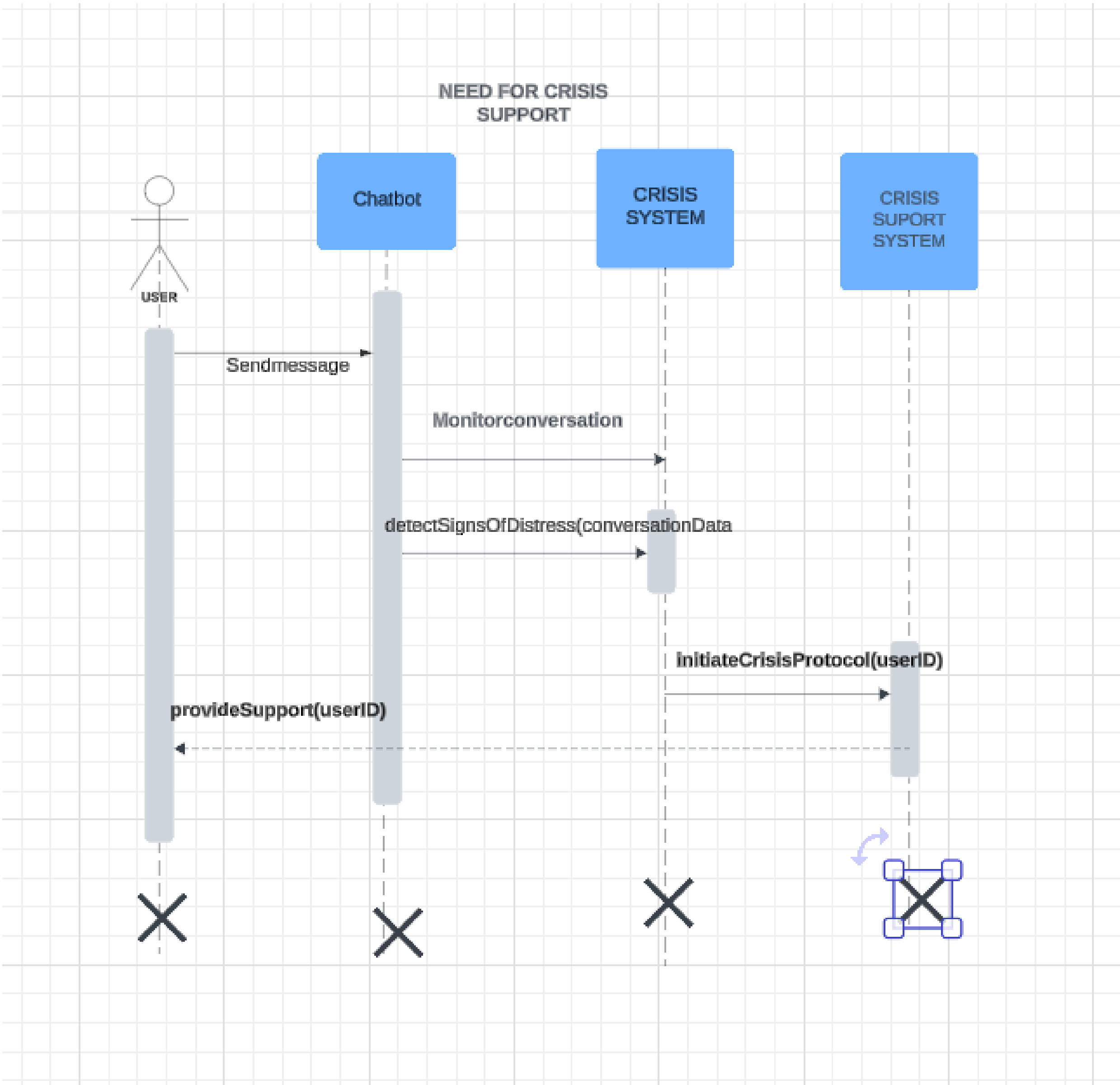
10. Moderate Users



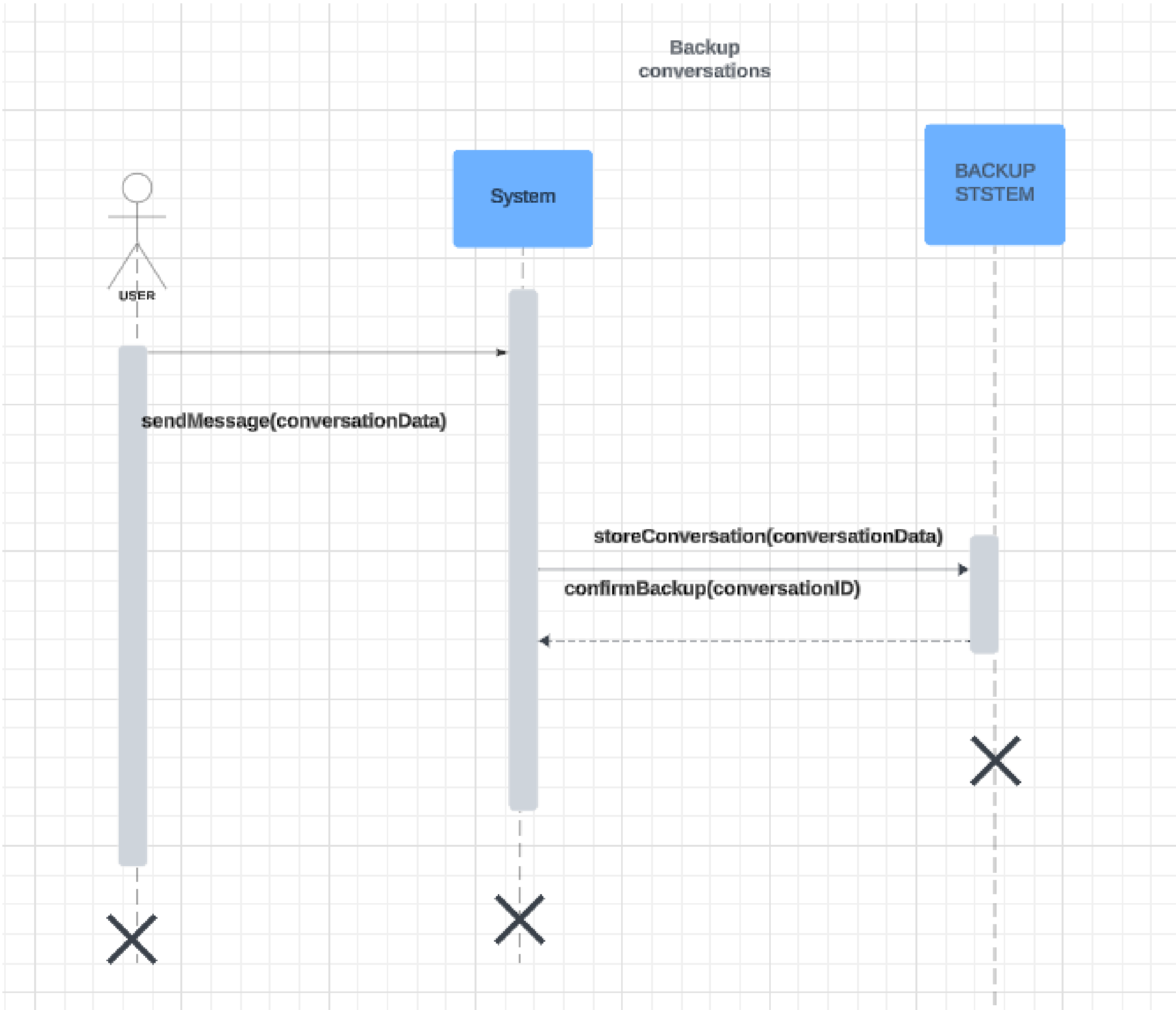
11. Answer Queries/Feedback



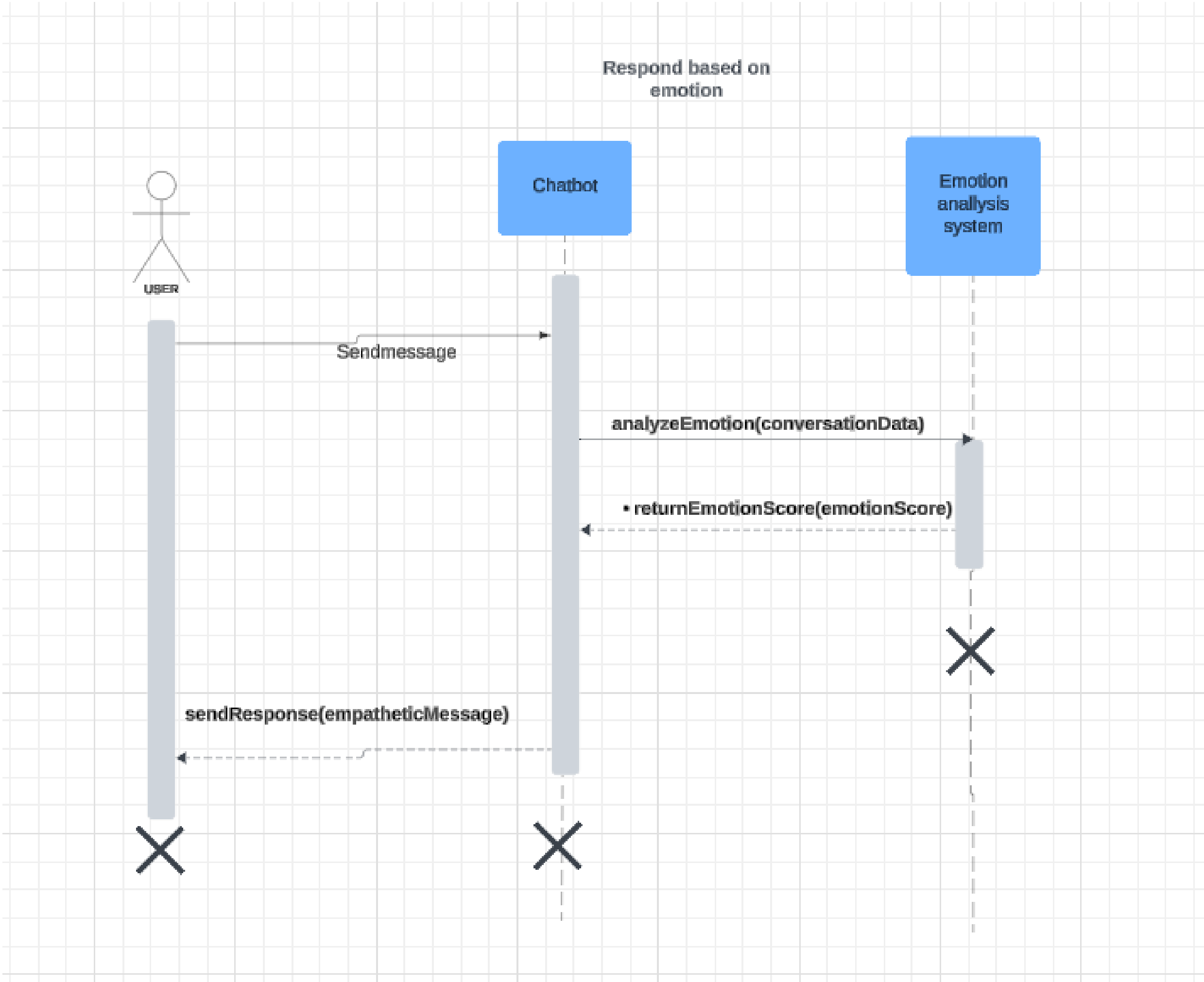
12. Evaluate Need for Crisis Support



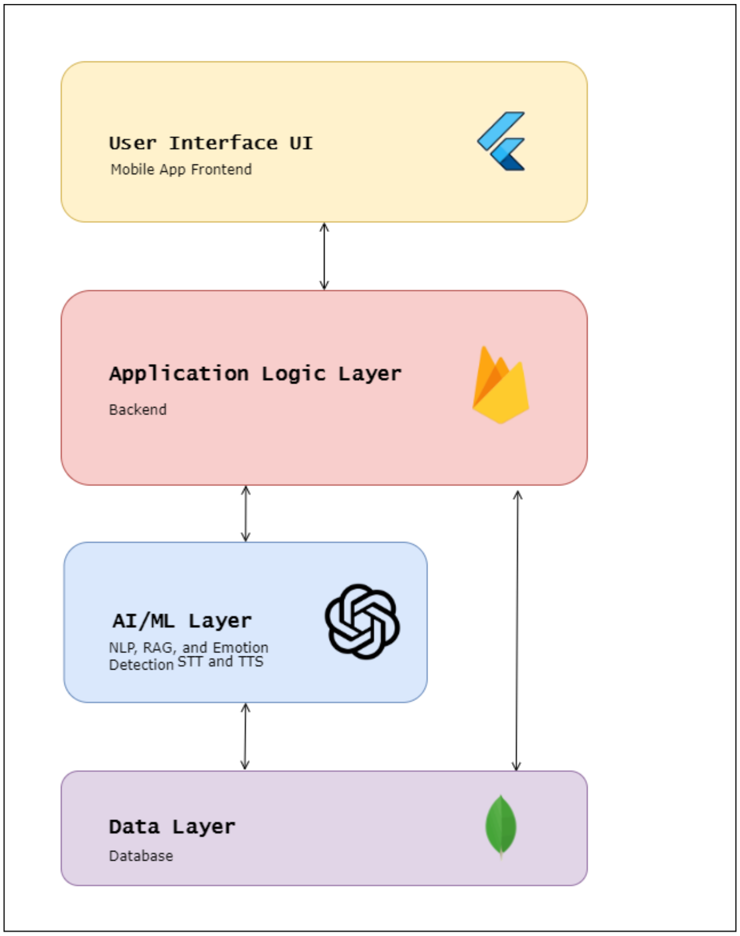
13. Backup Conversations



14. Respond Based on Emotion



Architecture Diagram:



The architecture we are going with is Layered. The system would compromise of 4 layers separated but would communicate with each other. This approach would guarantee security and make the system more compact.

The system has following layers:

User Interface:

- User Interface to get input from the user.

Application Logic Layer:

- This would contain the business logic of the system with other engines for emotion detection/recognition and recommendation system.

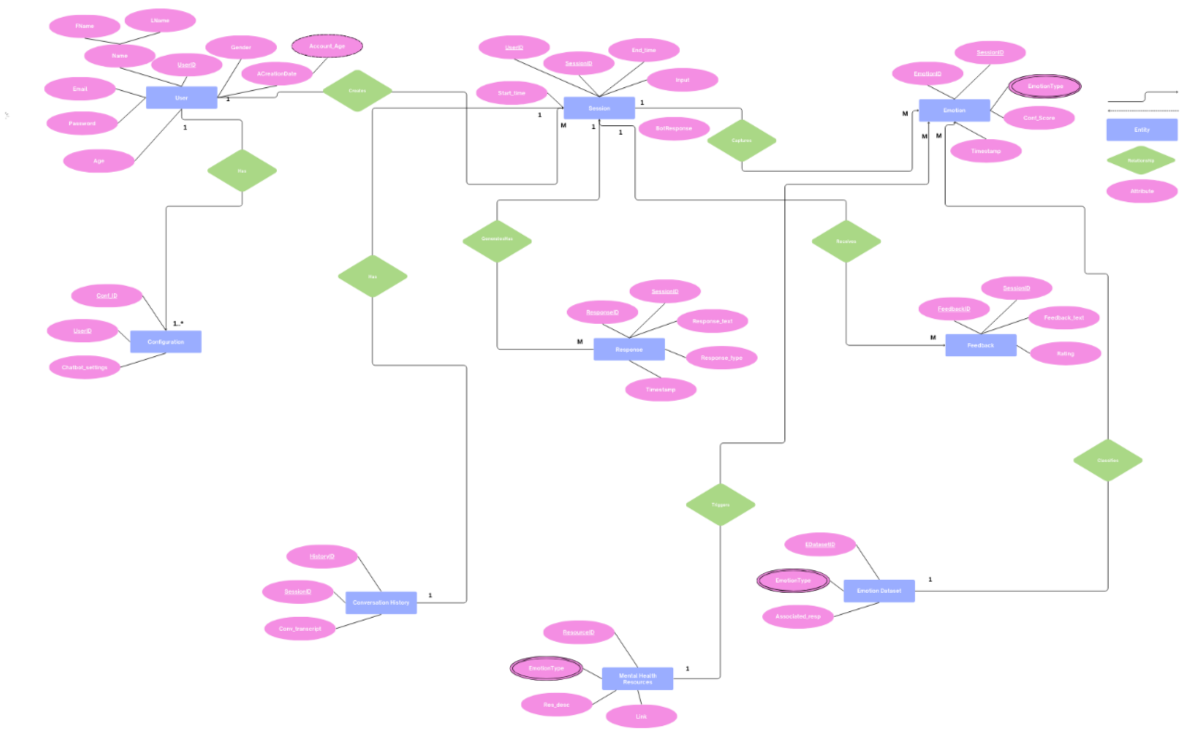
AI/ML Layer:

- This layer would contain the api for external components such as LLM.

Data Layer:

- This layer would contain the database of the system.

Entity Relation Diagram (ERD):

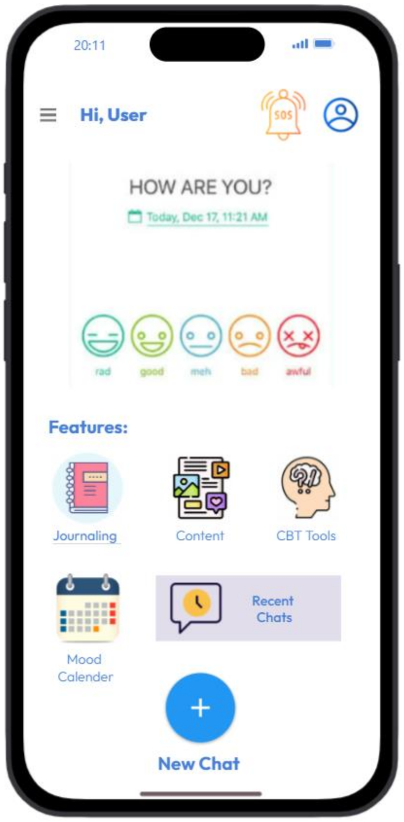


This is the Entity Relation Diagram of the system to create the relational schema for the creation of the Database for the system. This is the initial design and could change with the progress of the system.

- The oval shapes are the attributes of the entities.
- The squares are the entities.
- Diamonds are the relationships between different entities.

Prototype:

These are our initial designs for our app UI. This is just to give you an idea of what we are trying to achieve here.



Conclusion:

Our application will be designed to help people with their mental health by giving them responses according to the emotional state of the users. This the SRS documentation of our app. It includes the diagram necessary like activity, SDs and SSDs that are required for the documents. After this we will move to our 2nd phase in which we would start working on our STT and response modules with UI as well.